WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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Abstract.

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1. Introduction

1.1. Background and motivation. [Zhijie will write the background in the Dunkl setting.]

Theorem 1.1. *Let R be a normalised root system in $\mathbb{R}^N$ and let G be the corresponding finite reflection group. Let $k : R \to \mathbb{R}_+$ be G-invariant multiplicity function. Let $R_{\mathrm{Dunkl},j}$ be the j-th Dunkl-Riesz transform corresponding to R and k. Let w be the Dunkl weight corresponding to R and k. Let $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N)$.*

(i) *If $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$, $p > N$, then $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$. Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))} \leq c_{p,R,k}\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

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(ii) *If* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$, $p > N$, *then* $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$. *Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N,w(x)dx))} \le c_{p,R,k}^{-1}\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

(iii) *If* $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, *then* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. *Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N,w(x)dx))} \le c_{R,k}\|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

(iv) *If* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, *then* $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. *Furthermore,*

$$\|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N,w(x)dx))} \ge c_{R,k}^{-1}\|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

1.2. **Notation.** Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N, p, and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \le CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda\,\ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \le p \le \infty$, we denote by p' the conjugate of p, which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a μ-measurable set E by

$$(f)_E := \fint_E f(x)d\mu(x) := \frac{1}{\mu(E)}\int_E f(x)d\mu(x). \tag{1.1}$$

For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

2. Preliminaries

2.1. **Harmonic analysis in the Dunkl setting.** Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle\cdot,\cdot\rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane orthogonal to a normalised vector v is defined by

$$\sigma_v x := x - 2\langle x, v\rangle v.$$

A finite set $R \subset \mathbb{R}^N\backslash\{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $|v| = 1$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v\in R}$ is called the Weyl group (reflection group) of the root system R. Let $k : R \to [0, \infty)$ be a multiplicity function, which is a G-invariant function fixed throughout the paper.

Define the Dunkl measure $w(x)dx$ on $\mathbb{R}^N$ by

$$w(x) = \prod_{v\in R} |\langle x, v\rangle|^{k(v)}.$$

Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v\in R} k(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N,\ t, r > 0.$$

Furthermore,

$$w(B(x,r)) \sim r^N \prod_{v\in R} (|\langle x,v\rangle| + r)^{k(v)}, \tag{2.1}$$

where the implicit constant only depends on N, R and k. It follows from (2.1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$C^{-1}\left(\frac{r_2}{r_1}\right)^N \leq \frac{w(B(x,r_2))}{w(B(x,r_1))} \leq C\left(\frac{r_2}{r_1}\right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2, \tag{2.2}$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v\in R} \frac{k(v)}{2}\langle v,\xi\rangle \frac{f(x) - f(\sigma_v(x))}{\langle v,x\rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ. Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v\in R} k(v)\delta_v f(x),$$

where $\Delta_{\mathbb{R}^N}$ is the Laplacian on $\mathbb{R}^N$ and δ_v is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v,x\rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v,x\rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} := T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$, $j = 1, 2, \ldots, N$.

Definition 2.1. The Weyl chambers are defined as the closures of connected components of the set

$$\left\{x \in \mathbb{R}^N : \langle x,v\rangle \neq 0 \quad \textit{for all } v \in R\right\}. \tag{2.3}$$

Define

$$d(x,y) = \min_{\sigma\in G} |x - \sigma(y)|, \quad x,y \in \mathbb{R}^N.$$

Recall from [7, Chapter VII, proof of Theorem 2.12] that

$$d(x,y) = \|x-y\| \text{ if and only if } x,y \in \Gamma \tag{2.4}$$

for some closed Weyl chamber Γ.

Denote by $K_{\text{Dunkl},j}(x,y)$ the integral kernel of $R_{\text{Dunkl},j}$. We recall the kernel estimates for the Riesz transforms in the Dunkl setting proved in [5]. Although these estimates may not be sharp (see the comment following Theorem 2.1 in [5, p. 10]), they suffice for our purposes.

Lemma 2.2. *There exists a constant C such that for $j \in \{1, 2, \cdots, N\}$ and every x, y with $d(x,y) \neq 0$,*

$$|K_{\text{Dunkl},j}(x,y)| \leq \frac{d(x,y)}{|x-y|} \cdot \frac{C}{w(B(x,d(x,y)))}.$$

Proof. The proof is given in [5, Theorem 1.1]. □

Lemma 2.3. *We have*

$$d(x,y)^2 \cdot w(x)w(y) \le Cw^2(B(x,d(x,y))), \quad x,y \in \mathbb{R}^N.$$

Proof. Define equivalence relation $\sim$ on R by saying $v_1 \sim v_2$ if there exists $g \in G$ with $v_1 = gv_2$. Let us split R into equivalence classes with respect to $\sim$. Since k is G-invariant, it follows that k is constant on each equivalence class. Note that G leaves each equivalence class invariant.

We, therefore, write

$$w(x) = \prod_{A \in R/\sim} \big(\prod_{v \in A} |\langle x, v\rangle| \big)^{k|_A}.$$

On the other hand,

$$w(B(x,r)) \ge c^{(1)}_{R,k} r^N \prod_{A \in R/\sim} \big(\prod_{v \in A} (r + |\langle x, v\rangle|) \big)^{k|_A}.$$

Let $x \in \Gamma$ and $y \in g\Gamma$. Set $z = g^{-1}y \in \Gamma$. Note that $d(x,y) = |x-z|$ and, therefore,

$$|\langle x, v\rangle| + d(x,y) = |\langle x, v\rangle| + |x-z| \ge 2^{-\frac{1}{2}} |\langle z, v\rangle|.$$

Thus,

$$\prod_{v \in A} (|\langle x, v\rangle| + d(x,y)) \ge \prod_{v \in A} 2^{-\frac{1}{2}} |\langle z, v\rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle g^{-1}y, v\rangle| =$$

$$= 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, gv\rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in gA} |\langle y, v\rangle| = 2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, v\rangle|.$$

Thus,

$$w(B(x,d(x,y))) \ge c^{(1)}_{R,k} d^N(x,y) \prod_{A \in R/\sim} \big(\prod_{v \in A} (d(x,y) + |\langle x, v\rangle|) \big)^{k|_A} \ge$$

$$\ge c^{(1)}_{R,k} d^N(x,y) \prod_{A \in R/\sim} \big(2^{-\frac{1}{2}|A|} \prod_{v \in A} |\langle y, v\rangle| \big)^{k|_A} = c^{(2)}_{R,k} d^N(x,y) w(y).$$

On the other hand,

$$w(B(x,d(x,y))) \ge c^{(1)}_{R,k} d^N(x,y) w(x).$$

Multiplying the last two equations, we complete the proof. □

Lemma 2.4. *Let $1 \le j \le N$. If $x_j \ge y_j$, then*

$$K_{\mathrm{Dunkl},j}(x,y) \ge c_{R,k} \frac{x_j - y_j}{|x-y|} \cdot \frac{1}{w(B(x,|x-y|))}.$$

If $x_j \le y_j$, then

$$K_{\mathrm{Dunkl},j}(x,y) \le c_{R,k} \frac{x_j - y_j}{|x-y|} \cdot \frac{1}{w(B(x,|x-y|))}.$$

Proof. The lower bound is identical to that in [5, Theorem 1.2]. □

2.2. **Schatten class.** [Zhijie will add this later]

2.3. **Nearly weakly orthonormal sequences.** Rochberg and Semmes [16] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [4, 9, 15, 2, 3, 13, 16, 19, 18]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [9, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N, \|\cdot\|, w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Let $\mathscr{D}$ be a system of dyadic cubes. A sequence of functions $\{e_Q\}_{Q\in\mathscr{D}}$ is an NWO sequence if

(1) $\operatorname{supp} e_Q \subseteq cQ$ *for some positive constant c independent of Q*
(2) $\|e_Q\|_{L^s(\mathbb{R}^N,dw)} \le w(Q)^{1/s-1/2}$ *for some $s>2$.*

Lemma 2.5. *Let $1<p<\infty$, and let $\mathscr{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. For any compact operator T on $L^2(\mathbb{R}^N, dw)$, we have*

$$\Big(\sum_{Q\in\mathscr{D}} |\langle Te_Q, f_Q\rangle_{L^2(\mathbb{R}^N,dw)}|^p\Big)^{1/p} \le C_{N,p,w}\|T\|_{S^p},$$

where $\{e_Q\}_{Q\in\mathscr{D}}$ and $\{f_Q\}_{Q\in\mathscr{D}}$ are NWO sequences.

Proof. This property can be found in [16, (1.10)] (see also [9, Section 7]). □

Lemma 2.6. *Let $0<p<\infty$, and let $\mathscr{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. Assume that*

$$A = \sum_{Q\in\mathscr{D}} \lambda_Q \langle\cdot,\, e_Q\rangle_{L^2(\mathbb{R}^N,dw)} f_Q,$$

where $\{e_Q\}_{Q\in\mathscr{D}}$ and $\{f_Q\}_{Q\in\mathscr{D}}$ are NWO sequences and $\{\lambda_Q\}_{Q\in\mathscr{D}}$ is a sequence of scalars. We have

$$\|A\|_{S^p} \le C_{N,p,w}\|\{\lambda_Q\}_Q\|_{\ell^p}. \tag{2.5}$$

Proof. This property can be found in [16, (1.9)] (see also [9, Section 7]). □

3. Comparison of Sobolev norms

Theorem 3.1. *For every $f\in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$, $p>N$, we have*

$$\Big(\int_{\mathbb{R}^N\times\mathbb{R}^N} \Big(\frac{d(x,y)}{|x-y|}\Big)^p |f(x)-f(y)|^p \frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}} \le c_{p,R}\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Lemma 3.2. *For every convex polyhedral cone Γ, we have*

$$\int_\Gamma \frac{dx}{|x-y|^{2N}} \ge c_\Gamma \operatorname{dist}(y,\Gamma)^{-N}, \quad y\notin\Gamma.$$

Proof. Find a point $z\in\Gamma$ such that

$$|y-z| = \operatorname{dist}(y,\Gamma).$$

We have $z+\Gamma\subset\Gamma$. Thus,

$$\int_\Gamma \frac{dx}{|x-y|^{2N}} \ge \int_{z+\Gamma} \frac{dx}{|x-y|^{2N}} = \int_\Gamma \frac{dx}{|x-(y-z)|^{2N}}.$$

Clearly,

$$|x-(y-z)| \le |x| + |y-z|.$$

Thus,

$$\int_\Gamma \frac{dx}{|x-y|^{2N}} \ge \int_\Gamma \frac{dx}{(|x|+|y-z|)^{2N}} = |y-z|^{-N} \int_\Gamma \frac{dx}{(|x|+1)^{2N}}.$$

□

Lemma 3.3. *If $p \in (N, 2N)$, then*

$$\sup_{a>0} \int_{\mathbb{R}^N} \frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p} < \infty.$$

Proof. **Step 1:** we claim that, for every $a > 0$,

$$\int_{\mathbb{R}^N} \frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p} \le$$

$$\le 4^p \int_{\mathbb{R}^{N-1}\times(-a,a)} \frac{|x|^{p-2N}dx}{(1+|x'|+a)^p} + 3^{2N-p} \int_{\mathbb{R}^N} \frac{|x|^{p-2N}dx}{(1+|x|)^p}.$$

We clearly have

$$\int_{\mathbb{R}^N} \frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p} =$$

$$= \int_{\mathbb{R}^{N-1}\times(\frac{a}{2},\frac{3a}{2})} \frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p} + \int_{\mathbb{R}^{N-1}\times\mathbb{R}\setminus(\frac{a}{2},\frac{3a}{2})} \frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p}.$$

If $x_N \notin (\frac{a}{2}, \frac{3a}{2})$, then $|x_N - a| \ge \frac{1}{3}|x_N|$. Hence, $|x - ae_N| \ge \frac{1}{3}|x|$. Thus,

$$\int_{\mathbb{R}^{N-1}\times\mathbb{R}\setminus(\frac{a}{2},\frac{3a}{2})} \frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p} \le$$

$$\le \int_{\mathbb{R}^{N-1}\times\mathbb{R}\setminus(\frac{a}{2},\frac{3a}{2})} \frac{3^{2N-p}|x|^{p-2N}dx}{(1+|x|)^p} \le \int_{\mathbb{R}^N} \frac{3^{2N-p}|x|^{p-2N}dx}{(1+|x|)^p}.$$

If $x_N \in (\frac{a}{2}, \frac{3a}{2})$, then (here, we using the notation $x' = (x_1, \cdots, x_{N-1})$)

$$1+|x| \ge 1 + \max\{|x'|, \frac{1}{2}a\} \ge \frac{1}{4}(1+|x'|+a).$$

Thus,

$$\int_{\mathbb{R}^{N-1}\times(\frac{a}{2},\frac{3a}{2})} \frac{|x-ae_N|^{p-2N}dx}{(1+|x|)^p} \le 4^p \int_{\mathbb{R}^{N-1}\times(\frac{a}{2},\frac{3a}{2})} \frac{|x-ae_N|^{p-2N}dx}{(1+|x'|+a)^p} =$$

$$= 4^p \int_{\mathbb{R}^{N-1}\times(-\frac{a}{2},\frac{a}{2})} \frac{|x|^{p-2N}dx}{(1+|x'|+a)^p} \le 4^p \int_{\mathbb{R}^{N-1}\times(-a,a)} \frac{|x|^{p-2N}dx}{(1+|x'|+a)^p}.$$

Combining all preceding estimates, we infer the claim in Step 1.

Step 2: we claim that

$$\sup_{a>0} \int_{\mathbb{R}^{N-1}\times(-a,a)} \frac{|x|^{p-2N}dx}{(1+|x'|+a)^p} < \infty.$$

Passing to polar coordinates in $\mathbb{R}^{N-1}$, we equivalently rewrite the claim as

$$\sup_{a>0} \int_{\mathbb{R}_+\times(0,a)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(1+r+a)^p} < \infty.$$

If $0 < a < 1$, then

$$\int_{\mathbb{R}_+\times(0,a)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(1+r+a)^p} \le \int_{\mathbb{R}_+\times(0,1)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(1+r)^p}.$$

If $a \ge 1$, then

$$\int_{\mathbb{R}_+\times(0,a)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(1+r+a)^p} \le \int_{\mathbb{R}_+\times(0,a)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(r+a)^p} =$$

$$= \int_{(a,\infty)\times(0,a)} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(r+a)^p} +$$

$$+ \int_0^a \int_0^r \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dx_N dr}{(r+a)^p} + \int_0^a \int_0^{x_N} \frac{(r^2+x_N^2)^{\frac{p}{2}-N} r^{N-2} dr dx_N}{(r+a)^p} \le$$

$$\le {}^{1} \int_{(a,\infty)\times(0,a)} \frac{r^{p-2N} r^{N-2} dr dx_N}{r^p} +$$

$$+ \int_0^a \int_0^r \frac{r^{p-2N} r^{N-2} dx_N dr}{a^p} + \int_0^a \int_0^{x_N} \frac{x_N^{p-2N} r^{N-2} dr dx_N}{a^p} =$$

$$= \int_a^\infty r^{-N-1} dr + a^{-p} \int_0^a r^{p-N-1} dr + \frac{1}{N-1} a^{-p} \int_0^a x_N^{p-N-1} dx_N =$$

$$= \frac{1}{N} a^{-N} + \frac{1}{p-N} a^{-N} + \frac{1}{(N-1)(p-N)} a^{-N} \le \frac{1}{N} + \frac{1}{p-N} + \frac{1}{(N-1)(p-N)}.$$

This yields the claim in Step 2.

Combining Steps 1 and 2, we complete the proof of the lemma. □

Lemma 3.4. *If $p \in (N, 2N)$, then*

$$\int_{\mathbb{R}^{N-1}\times\mathbb{R}_+} \frac{|x-y|^{p-2N} dx}{|x+z|^p} \le c_{p,N} z_N^{-N}, \quad y, z \in \mathbb{R}^{N-1}\times\mathbb{R}_+.$$

Proof. By dilation invariance, we may assume without loss of generality that $z_N = 1$. By translation invariance in $\mathbb{R}^{N-1}$, we may assume without loss of generality that $z_1 = \cdots = z_{N-1} = 0$. We, therefore, have

$$|x+z| = |x+e_N| \ge \max\{1, |x|\} \ge \frac{1}{2}(1+|x|), \quad x \in \mathbb{R}^{N-1}\times\mathbb{R}_+.$$

Thus,

$$\int_{\mathbb{R}^{N-1}\times\mathbb{R}_+} \frac{|x-y|^{p-2N} dx}{|x+z|^p} \le 2^p \int_{\mathbb{R}^{N-1}\times\mathbb{R}_+} \frac{|x-y|^{p-2N} dx}{(1+|x|)^p} \le$$

$$= 2^p \int_{\mathbb{R}^N} \frac{|x-y|^{p-2N} dx}{(1+|x|)^p} = 2^p \int_{\mathbb{R}^N} \frac{\big|x-|y|e_N\big|^{p-2N} dx}{(1+|x|)^p}.$$

The assertion follows from Lemma 3.3. □

We need the following elementary version of the Hahn-Banach theorem.

Lemma 3.5. *Let $K \subset \mathbb{R}^N$ be a convex set and let $y \notin K$. Let $z \in K$ be the point closest to y. Affine hyperplane P passing through z and orthogonal to $y - z$ splits the space into two half-spaces, one of them containing y, the other one containing K.*

[1] It is exactly at this point we are using the assumption $p < 2N$.

Proof. By translation invariance, we may assume without loss of generality that $z = 0$. By rotation and dilation invariance, we may assume without loss of generality that $y = e_N$. By assumption, $\mathrm{dist}(e_N, K) = 1$. We need to show that K is contained in the (closed) lower half-plane.

Assume the contrary: let $x \in K$ be such that $x_N > 0$. Since $0 \in K$ and since K is convex, it follows that $\epsilon x \in K$ for every $\epsilon \in (0,1)$. Thus,

$$\mathrm{dist}(e_N, K) \leq |e_N - \epsilon x|, \quad \epsilon \in (0,1).$$

Since $\epsilon \in (0,1)$ is arbitrary, it follows that

$$\mathrm{dist}(e_N, K) \leq \inf_{\epsilon \in (0,1)} |e_N - \epsilon x| = \inf \big(\epsilon^2 \sum_{k=1}^{N-1} x_k^2 + (1 - \epsilon x_N)^2\big)^{\frac{1}{2}}.$$

Since $x_N > 0$, it follows that infimum is strictly less than 1. Thus, $\mathrm{dist}(e_N, K) < 1$. However, it is granted in the lemma that $\mathrm{dist}(e_N, K) = 1$. This contradiction shows that such an x cannot exist. This completes the proof. □

Lemma 3.6. *If $p \in (N, 2N)$, then*

$$\int_\Gamma \frac{|x-y|^{p-2N} dx}{|x - gy|^p} \leq c_p \mathrm{dist}(gy, \Gamma)^{-N}, \quad y \in \Gamma.$$

Proof. Find a point $z \in \Gamma$ such that

$$|gy - z| = \mathrm{dist}(gy, \Gamma).$$

By Lemma 3.5, affine hyperplane passing through z and orthogonal to $gy - z$ splits the space into two half-spaces, one of them containing gy, the other one containing Γ (denote this one by H).

We have $y \in H$, $\Gamma \subset H$ and $gy \notin H$. So,

$$\int_\Gamma \frac{|x-y|^{p-2N} dx}{|x - gy|^p} \leq \int_H \frac{|x-y|^{p-2N} dx}{|x - gy|^p}.$$

Using Lemma 3.4, we write

$$\int_H \frac{|x-y|^{p-2N} dx}{|x - gy|^p} \leq c_{p,N} \mathrm{dist}(gy, H)^{-N}.$$

Since $z \in \partial H$ and since $gy - z$ is orthogonal to ∂H, it follows that $\mathrm{dist}(gy, H) = |gy - z| = \mathrm{dist}(gy, \Gamma)$. □

Proof of Theorem 3.1. **Case 1:** Let $p \in (N, 2N)$.

We have

$$\Big(\int_{\Gamma \times g\Gamma} \Big(\frac{d(x,y)}{|x-y|}\Big)^p |f(x) - f(y)|^p \frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}} =$$

$$= \Big(\int_{\Gamma \times \Gamma} \Big(\frac{|x-y|}{|x-gy|}\Big)^p |f(x) - f(gy)|^p \frac{dxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}} \leq$$

$$\leq \Big(\int_{\Gamma \times \Gamma} \Big(\frac{|x-y|}{|x-gy|}\Big)^p |f(x) - f(y)|^p \frac{dxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}} +$$

$$+ \Big(\int_{\Gamma \times \Gamma} \Big(\frac{|x-y|}{|x-gy|}\Big)^p |f(y) - f(gy)|^p \frac{dxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}} \leq$$

$$\leq \Big(\int_{\Gamma \times \Gamma} |f(x) - f(y)|^p \frac{dxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}} +$$

$$+\Big(\int_\Gamma |f(y)-f(gy)|^p\Big(\int_\Gamma \frac{|x-y|^{p-2N}}{|x-gy|^p}dx\Big)dy\Big)^{\frac{1}{p}}.$$

Obviously, the first summand on the right hand side is less than $\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}$. By Lemmas 3.6 and 3.2,

$$\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}+$$

$$+c^{(1)}_{p,R}\Big(\int_\Gamma |f(y)-f(gy)|^p\Big(\int_\Gamma \frac{dx}{|x-gy|^{2N}}\Big)dy\Big)^{\frac{1}{p}}=$$

$$=\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}+c^{\frac{1}{p}}_{p,R}\Big(\int_{\Gamma\times\Gamma}\frac{|f(y)-f(gy)|^p dxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}\leq$$

$$\leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}+c^{(1)}_{p,R}\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(gy)|^p dxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}+$$

$$+c^{(1)}_{p,R}\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^p dxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}.$$

Now,

$$\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(gy)|^p dxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}=\Big(\int_{\Gamma\times g\Gamma}\frac{|f(x)-f(y)|^p dxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}}\leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}$$

and

$$\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^p dxdy}{|x-gy|^{2N}}\Big)^{\frac{1}{p}}\leq\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^p dxdy}{|x-y|^{2N}}\Big)^{\frac{1}{p}}\leq \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Thus,

$$\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq (1+2c^{(1)}_{p,R})\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Summing over Γ and $1\neq g\in G$, we obtain

$$\Big(\int_{\mathbb{R}^N\times\mathbb{R}^N}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq |G|^2(1+2c^{(1)}_{p,R})\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Case 2: Let $p\geq 2N$. We have

$$\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^p|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}\leq$$

$$\leq\Big(\int_{\Gamma\times g\Gamma}\Big(\frac{d(x,y)}{|x-y|}\Big)^{2N}|f(x)-f(y)|^p\frac{dxdy}{d(x,y)^{2N}}\Big)^{\frac{1}{p}}=\|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

□

4. Proof of Theorem 1.1 (i)

Let (X,μ) be a measure space and let $L_p(L_{q,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)$ denote the mixed-norm space consisting of all $\mu\times\mu$-measurable functions $f: X\times X\to\mathbb{C}$ such that

$$\int_X \|f(x,\cdot)\|^p_{L_{q,\infty}(X,\mu)}d\mu(x), \int_X \|f(\cdot,y)\|^p_{L_{q,\infty}(X,\mu)}d\mu(y)<\infty.$$

We equip $L_p(L_{q,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)$ with its natural norm.

Lemma 4.1. *Let (X,μ) be a measure space and let $2<p<\infty$. Suppose that $K\in L_p(L_{p',\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)$. The integral operator*

$$T_K f(x):=\int_X K(x,y)f(y)d\mu(y)$$

belongs to $S_{p,\infty}(L_2(X,d\mu))$. Moreover, we have

$$\|T_K\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|K\|_{L_p(L_{p',\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}.$$

Proof. The proof is given in [11, Lemma 2] (see also [9, Corollary 11.2]). □

Lemma 4.2. *Let $p>2$ and let $\frac{1}{q}=1-\frac{2}{p}$. We have*

$$\|F_1F_2\|_{L_p(L_{p',\infty})(X\times X,\mu\times\mu)}\le 2^{\frac{1}{pp'}}\|F_1\|_{L_p(X\times X,\mu\times\mu)}\|F_2\|_{L_\infty(L_{q,\infty})(X\times X,\mu\times\mu)}.$$

Proof. We clearly have

$$LHS=\Big(\int_X \|F_1(x,\cdot)F_2(x,\cdot)\|^p_{L_{p',\infty}(X,\mu)}d\mu(x)\Big)^{\frac{1}{p}}.$$

Clearly, $\frac{1}{p'}=\frac{1}{p}+\frac{1}{q}$. Thus,

$$\|F_1(x,\cdot)F_2(x,\cdot)\|_{L_{p',\infty}(X,\mu)}\le 2^{\frac{1}{p'}}\|F_1(x,\cdot)\|_{L_{p,\infty}(X,\mu)}\|F_2(x,\cdot)\|_{L_{q,\infty}(X,\mu)}\le$$
$$\le 2^{\frac{1}{p'}}\|F_1(x,\cdot)\|_{L_p(X,\mu)}\|F_2(x,\cdot)\|_{L_{q,\infty}(X,\mu)}.$$

Hence,

$$LHS\le 2^{\frac{1}{pp'}}\Big(\int_X \|F_1(x,\cdot)\|^p_{L_p(X,\mu)}\|F_2(x,\cdot)\|^p_{L_{q,\infty}(X,\mu)}d\mu(x)\Big)^{\frac{1}{p}}\le$$
$$\le 2^{\frac{1}{pp'}}\Big(\int_X \|F_1(x,\cdot)\|^p_{L_p(X,\mu)}d\mu(x)\Big)^{\frac{1}{p}}\times\sup_{x\in X}\|F_2(x,\cdot)\|_{L_{q,\infty}(X,\mu)}=$$
$$=2^{\frac{1}{pp'}}\|F_1\|_{L_p(X\times X,\mu\times\mu)}\|F_2\|_{L_\infty(L_{q,\infty})(X\times X,\mu\times\mu)}.$$

□

Lemma 4.3. *Let (X,μ) be a measure space and let $2<p<\infty$. Suppose that $F\in L_p(X\times X,H^2(\mu\times\mu))$ and $H\in L_\infty(L_{1,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)$. We have*

$$\|T_{FH}\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\|H\|^{1-\frac{2}{p}}_{L_\infty(L_{1,\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}.$$

Proof. By Lemma 4.1,

$$\|T_{FH}\|_{S_{p,\infty}(L_2(X,d\mu))}\le c_p\|FH\|_{L_p(L_{p',\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}.$$

Let $\frac{1}{q}=1-\frac{2}{p}$. It follows from Lemma 4.2 that

$$\|F_1F_2\|_{L_p(L_{p',\infty})^{\mathrm{symm}}(X\times X,\mu\times\mu)}\le$$

$$\leq 2^{\frac{1}{pp'}}\|F_1\|_{L_p(X\times X,\mu\times\mu)}\|F_2\|_{L_\infty(L_{q,\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

Setting $F_1 = FH^{\frac{2}{p}}$ and $F_2 = H^{\frac{1}{q}}$, we obtain

$$\|FH\|_{L_p(L_{p',\infty})^{\rm symm}(X\times X,\mu\times\mu)} \leq$$

$$\leq 2^{\frac{1}{pp'}}\|FH^{\frac{2}{p}}\|_{L_p(X\times X,\mu\times\mu)}\|H^{\frac{1}{q}}\|_{L_\infty(L_{q,\infty})^{\rm symm}(X\times X,\mu\times\mu)} =$$

$$= 2^{\frac{1}{pp'}}\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\|H\|^{\frac{1}{q}}_{L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)} =$$

$$= 2^{\frac{1}{pp'}}\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}\|H\|^{1-\frac{2}{p}}_{L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)}.$$

□

Lemma 4.4. *Let (X,μ) be a measure space and let $2 < p < \infty$. Suppose that $F \in L_p(X\times X, GH^2(\mu\times\mu))$ and $H \in L_\infty(L_{1,\infty})^{\rm symm}(X\times X,\mu\times\mu)$. We have*

$$\|T_{FH}\|_{S_p(L_2(X,d\mu))} \leq c_{p,H}\|F\|_{L_p(X\times X,H^2(\mu\times\mu))}.$$

Proof. The assertion follows from Lemma 4.3 by real interpolation. □

The following proposition, strictly speaking, is not part of the proof of Theorem 1.1 (i). Instead, it features in the proof of Theorem 1.1 (ii) in the next section. However, in terms of the employed technique, it certainly belongs to this section.

Proposition 4.5. *Let $p > 2$, let Γ be a chamber and let $f \in \dot{W}^{\frac{N}{p},p}(\Gamma)$. Let V be an integral operator whose integral kernel K satisfies the condition*

$$|K(x,y)| \leq \frac{1}{w(B(x,|x-y|))}.$$

For every $1 \leq j \leq N$, we have

$$\|[V,M_f]\|_{\mathcal{L}_p(L_2(\Gamma,w(x)dx))} \leq c_{R,k}\|f\|_{\dot{W}^{\frac{N}{p},p}(\Gamma)}.$$

Proof. In what follows, let $X = \Gamma$ and $d\mu(x) = w(x)dx$. Set

$$F(x,y) = w(B(x,|x-y|))K(x,y)(f(x)-f(y)), \quad x \neq y,$$

$$H(x,y) = \frac{1}{w(B(x,|x-y|))}, \quad x \neq y.$$

By [9, Lemma 5.7], $H \in L_\infty(L_{1,\infty})^{\rm symm}(\Gamma\times\Gamma,\mu\times\mu)$. By assumption on the kernel K,

$$\|F\|_{L_p(\Gamma\times\Gamma,G^2(\mu\times\mu))} \leq \Big(\int_{\Gamma\times\Gamma}|f(x)-f(y)|^p\frac{w(x)w(y)dxdy}{w^2(B(x,|x-y|))}\Big)^{\frac{1}{p}}.$$

By (2.1),

$$w(B(x,|x-y|)) \geq c'_{R,k}|x-y|^N w^{\frac{1}{2}}(x)w^{\frac{1}{2}}(y), \quad x,y \in \mathbb{R}^N.$$

Thus,

$$\|F\|_{L_p(\Gamma\times\Gamma,H^2(\mu\times\mu))} \leq$$

$$\leq (c'_{R,k})^{-\frac{2}{p}}\Big(\int_{\Gamma\times\Gamma}\frac{|f(x)-f(y)|^p}{|x-y|^{2N}}dxdy\Big)^{\frac{1}{p}} = (c'_{R,k})^{-\frac{2}{p}}\|f\|_{\dot{W}^{\frac{N}{p},p}(\Gamma)}.$$

Obviously, the integral kernel of the operator $[V,M_f]$ is FH. The assertion follows now from Lemma 4.4. □

Proof of Theorem 1.1 (i). In what follows, let $X = \mathbb{R}^N$ and $d\mu(x) = w(x)dx$. Set

$$F(x,y) = w(B(x,d(x,y)))K_{\mathrm{Dunkl},j}(x,y)(f(x)-f(y)), \quad x \neq y,$$

$$H(x,y) = \frac{1}{w(B(x,d(x,y)))}, \quad x \neq y.$$

By [9, Lemma 11.7], $H \in L_\infty(L_{1,\infty})^{\mathrm{symm}}(X \times X, \mu \times \mu)$. By Lemma 2.2,

$$\|F\|_{L_p(X\times X, H^2(\mu\times\mu))} \leq$$

$$\leq c_{R,k}^{(1)} \left(\int_{\mathbb{R}^N \times \mathbb{R}^N} \left(\frac{d(x,y)}{|x-y|} \right)^p |f(x)-f(y)|^p \frac{w(x)w(y)dxdy}{w(B(x,d(x,y)))^2} \right)^{\frac{1}{p}}.$$

Using Lemma 2.3, we conclude that

$$\|F\|_{L_p(X\times X, H^2(\mu\times\mu))} \leq c_{R,k}^{(2)} \left(\int_{\mathbb{R}^N \times \mathbb{R}^N} \left(\frac{d(x,y)}{|x-y|} \right)^p |f(x)-f(y)|^p \frac{dxdy}{d(x,y)^{2N}} \right)^{\frac{1}{p}}.$$

Using Theorem 3.1, we conclude that

$$\|F\|_{L_p(X\times X, H^2(\mu\times\mu))} \leq c_{R,k} \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)}.$$

Obviously, the integral kernel of the operator $[R_{\mathrm{Dunkl},j}, M_f]$ is FH. The assertion follows now from Lemma 4.4. □

5. Expected dyadic result

I expect the following result to be true.

Theorem 5.1. *Let Q_0 be a fixed dyadic cube touching the plane $\{x_N = 0\}$. If*

$$\left\{ \frac{1}{|(c_Q)_N|^N} \int_Q |f|(x)dx \right\}_{\substack{Q\in\mathcal{D} \\ Q\subset Q_0 \\ |(c_Q)_N| \geq c\cdot s(Q)}} \in l_p$$

for some $c > 0$, then

$$\int_{Q_0} |x_N|^{-N} |f|^p(x)dx < \infty.$$

6. Simplified version of Zhijie's argument

Lemma 6.1. *Let $\mathcal{D}$ be the standard system of dyadic cubes. Let $O_v \in O(N,\mathbb{R})$ be such $O_v e_N = v$ and let $g \in G$. We have*

$$\left\| \left\{ w(O_v Q)^{-1} \langle T\chi_{gO_vQ}, \chi_{O_vQ} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \right\}_{Q\in\mathcal{D}} \right\|_{l_p(\mathcal{D})} \leq c_{g,v} \|T\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}.$$

Proof. to be done later. □

It is decidedly inconvenient that integral kernel K_j of the operator $R_{\mathrm{Dunkl},j}$ is not positive.

Lemma 6.2. *Integral kernel of $R_{\mathrm{Dunkl},j}$ has the same sign as $(x-y)_j$.*

Proof. This should follow from Lemma 3.3 in refer to Dziubanski-Hejna paper Ji refers to. □

Consider the operator B_j I do not know whether this operator is bounded whose integral kernel is given by the formula

$$(x,y) \to |K_j|(x,y), \quad x,y \in \mathbb{R}^N.$$

Lemma 6.3. *Suppose* $M_{f\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_{g\Gamma}} \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$. *It follows that*

$$M_{|f|\chi_\Gamma} B_j M_{\chi_{g\Gamma}} \in \mathcal{L}_p(L_2(\Gamma, w(x)dx))$$

Proof. Multiplying by $\mathrm{sgn}(f)$ on the left, we may assume without loss of generality that $f \geq 0$. By Lemma 6.2, $K_j(x,y)$ has the same sign as $(x-y)_j$. Thus,

$$M_{f\chi_\Gamma} B_j M_{\chi_{g\Gamma}} = \mathfrak{S}_{\mathrm{sgn}((x-y)_j)}(M_{f\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_{g\Gamma}}).$$

Schur multiplier $\mathfrak{S}_{\mathrm{sgn}((x-y)_j)}$ is easily seen to be bounded on $\mathcal{L}_p(L_2(\Gamma, w))$. This completes the proof. □

Let $g \in \Gamma$ be a reflection with respect to vector $v_0 \in R$.

Lemma 6.4. *Let* $Q_0 \in \mathcal{D}$ *be such that*

(i) $Q_0 \subset O_{v_0}^{-1}\Gamma$ *touches the plane* $\{x_N = 0\}$.
(ii) $O_{v_0}Q_0$ *is separated from any plane* $\{\langle x, v\rangle = 0\}$, $v_0 \neq v \in R$.

If $Q \in \mathcal{D}$ *is such that* $Q \subset Q_0$ *and*[2]

$$|(c_Q)_N| \geq 2N^{\frac{1}{2}} s(Q), \quad s(Q) \leq \mathrm{const}_{Q_0}.$$

(i) *We have*

$$|\langle O_{v_0} c_Q, v\rangle| \geq 2N^{\frac{1}{2}} s(Q), \quad |\langle g O_{v_0} c_Q, v\rangle| \geq 2N^{\frac{1}{2}} s(Q), \quad v \in \mathbb{R}.$$

(ii) *If* $x \in O_{v_0}Q$ *and* $y \in gO_{v_0}Q$, *then*

$$\frac{1}{2}|\langle O_{v_0} c_Q, v\rangle| \leq |\langle x, v\rangle| \leq \frac{3}{2}|\langle O_{v_0} c_Q, v\rangle|,$$
$$\frac{1}{2}|\langle g O_{v_0} c_Q, v\rangle| \leq |\langle y, v\rangle| \leq \frac{3}{2}|\langle g O_{v_0} c_Q, v\rangle|.$$

(iii) *If* $x \in O_{v_0}Q$ *and* $y \in gO_{v_0}Q$, *then*

$$c_k^{-1} w(O_{v_0} c_Q) \leq w(x), w(y) \leq c_k w(O_{v_0} c_Q).$$

(iv) *If* $x \in O_{v_0}Q$ *and* $y \in gO_{v_0}Q$, *then*

$$|(c_Q)_N| \leq |x-y| \leq 3|(c_Q)_N|.$$

(v) *If* $x \in O_{v_0}Q$ *and* $y \in gO_{v_0}Q$, *then*

$$c_k^{-1}|(c_Q)_N|^N \cdot w(O_{v_0} c_Q) \leq w(B(x, |x-y|)) \leq c_k |(c_Q)_N|^N \cdot w(O_{v_0} c_Q).$$

(vi) *We have*

$$c_k^{-1} m(Q) w(O_{v_0} Q) \leq w(O_{v_0} Q) \leq c_k m(Q) w(O_{v_0} Q).$$

Proof. Since $c_Q \in Q_0$, it follows that

$$|\langle O_{v_0} c_Q, v\rangle| \geq \inf_{z \in Q_0} |\langle O_{v_0} z, v| \geq 2N^{\frac{1}{2}} \mathrm{const}_{Q_0} \geq 2N^{\frac{1}{2}} s(Q), \quad v_0 \neq v \in R,$$

$$|\langle g O_{v_0} c_Q, v\rangle| \geq \inf_{z \in Q_0} |\langle g O_{v_0} z, v| \geq 2N^{\frac{1}{2}} \mathrm{const}_{Q_0} \geq 2N^{\frac{1}{2}} s(Q), \quad v_0 \neq v \in R.$$

On the other hand, we have

$$|\langle g O_{v_0} c_Q, v_0\rangle| = |\langle O_{v_0} c_Q, v_0\rangle| = |\langle O_{v_0} c_Q, O_{v_0} e_N\rangle| = |\langle c_Q, e_N\rangle| = |(c_Q)_N|.$$

[2]Here, we use the shorthand

$$\mathrm{const}_{Q_0} = \frac{1}{2N^{\frac{1}{2}}} \min\{ \inf_{\substack{x \in Q_0 \\ v_0 \neq v \in R}} |\langle x, v\rangle|, \inf_{\substack{x \in Q_0 \\ v_0 \neq v \in R}} |\langle gx, v\rangle| \}.$$

This yields (i).

If $x \in O_{v_0}Q$ and $y \in gO_{v_0}Q$, then

$$|\langle x, v\rangle| \geq |\langle O_{v_0}c_Q, v\rangle| - |\langle (x - O_{v_0}c_Q), v\rangle| \geq |\langle O_{v_0}c_Q, v\rangle| - |x - O_{v_0}c_Q| \geq$$

$$\geq |\langle O_{v_0}c_Q, v\rangle| - N^{\frac{1}{2}}s(Q) \geq \frac{1}{2}|\langle O_{v_0}c_Q, v\rangle|, \quad x \in O_vQ.$$

The other inequalities in (ii) are proved similarly.

The assertion (iii) follows immediately from (ii).

If $x \in O_{v_0}Q$ and $y \in gO_{v_0}Q$, then

$$|x - y| \leq |x - O_{v_0}c_Q| + |y - gO_{v_0}c_Q| + |O_{v_0}c_Q - gO_{v_0}c_Q| \leq$$

$$\leq 2N^{\frac{1}{2}}s(Q) + |O_{v_0}c_Q - (O_{v_0}c_Q - 2\langle O_{v_0}c_Q, v_0\rangle v_0)| =$$

$$= 2N^{\frac{1}{2}}s(Q) + 2|\langle O_{v_0}c_Q, v_0\rangle| \leq 3|\langle O_{v_0}c_Q, v_0\rangle| = 3|(c_Q)_N|.$$

The other inequality in (iv) is proved similarly.

Note that

$$w(B(x, |x - y|)) \geq c_k^{-1}|x - y|^N \cdot w(x).$$

If $x \in O_{v_0}Q$ and $y \in gO_{v_0}Q$, then the it follows from (iii) and (iv) that

$$w(B(x, |x - y|)) \geq c_k^{-1}|\langle O_{v_0}c_Q, v_0\rangle|^N \cdot w(O_{v_0}c_Q).$$

Thus, we obtained the left hand side estimate in (v).

Note that

$$w(B(x, |x - y|)) \leq c_k|x - y|^N \cdot \prod_{v \in R}(|x - y| + |\langle x, v\rangle|)^{k_v}.$$

If $x \in O_{v_0}Q$ and $y \in gO_{v_0}Q$, then it follows from (ii) and (iv) that

$$w(B(x, |x - y|)) \leq c_k|\langle O_{v_0}c_Q, v_0\rangle|^N \cdot \prod_{v \in R}(|\langle O_{v_0}c_Q, v_0\rangle| + |\langle O_{v_0}c_Q, v\rangle|)^{k_v}.$$

By assumption on Q_0,

$$|\langle z, v_0\rangle| \leq \mathrm{const}'_{Q_0}|\langle z, v\rangle|, \quad z \in Q_0, \quad v \in R.$$

Since $O_{v_0}c_Q \in Q_0$, it follows that

$$|\langle O_{v_0}c_Q, v_0\rangle| \leq \mathrm{const}'_{Q_0}|\langle O_{v_0}c_Q, v\rangle|, \quad v \in R.$$

Hence,

$$w(B(x, |x - y|)) \leq c_k|\langle O_{v_0}c_Q, v_0\rangle|^N \cdot \prod_{v \in R}((1 + \mathrm{const}'_{Q_0})|\langle O_{v_0}c_Q, v\rangle|)^{k_v}.$$

Thus, we obtained the right hand side estimate in (v).

Finally, the assertion (vi) follows immediately from (iii). □

Lemma 6.5. *Suppose that v_0 is not orthogonal to e_j. Suppose $Q \in \mathcal{D}$ is such that*

$$|(c_Q)_N| \cdot |(v_0)_j| \geq 2N^{\frac{1}{2}}s(Q).$$

We have

$$\int_{gO_{v_0}Q} |(x - y)_j|dy \geq m(Q) \cdot |(c_Q)_N| \cdot |(v_0)_j|.$$

Proof. Clearly,

$$\int_{gO_{v_0}Q} |(x-y)_j| dy \geq \max\Big\{ \int_{gO_{v_0}Q} (x-y)_j dy, \int_{gO_{v_0}Q} (y-x)_j dy \Big\}.$$

Next,

$$\int_{gO_{v_0}Q} (x-y)_j dy = (x - gO_{v_0}c_Q)_j \cdot m(Q)$$

and

$$\begin{aligned} |(x-gO_{v_0}c_Q)_j| &\geq |(O_{v_0}c_Q)_j - (gO_{v_0}c_Q)_j| - |(x - O_{v_0c_Q})_j| \geq \\ &\geq |(O_{v_0}c_Q - gO_{v_0}c_Q)_j| - N^{\frac{1}{2}} s(Q) = \\ &= 2|\langle O_{v_0}c_Q, v_0\rangle| \cdot |(v_0)_j| - N^{\frac{1}{2}} s(Q) \geq |\langle O_{v_0}c_Q, v_0\rangle| \cdot |(v_0)_j|. \end{aligned}$$

Combining these inequalities, we complete the proof. □

Lemma 6.6. *Suppose $M_{f\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_{g\Gamma}} \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$. Let Q_0 be as in Lemma 6.4. If v_0 is not orthogonal to e_j, then*

$$\Big\{ \frac{1}{|(c_Q)_N|^N} \int_{O_{v_0}Q} |f|(x)dx \Big\}_{\substack{Q\in\mathcal{D} \\ Q\subset Q_0 \\ |(c_Q)_N| \geq c\cdot s(Q)}} \in l_p, \quad c = \frac{2N^{\frac{1}{2}}}{|(v_0)_j|}.$$

Proof. Combining Lemma 6.1 and Lemma 6.3, we obtain

$$\Big\{ w(O_{v_0}Q)^{-1} \langle |f| \cdot B_j(\chi_{gO_{v_0}Q}), \chi_{O_{v_0}Q}\rangle_{L_2(\mathbb{R}^N, w(x)dx)} \Big\}_{\substack{Q\in\mathcal{D} \\ O_{v_0}Q\subset\Gamma}} \in l_p.$$

In particular,

$$\Big\{ w(O_{v_0}Q)^{-1} \langle |f| \cdot B_j(\chi_{gO_{v_0}Q}), \chi_{O_{v_0}Q}\rangle_{L_2(\mathbb{R}^N, w(x)dx)} \Big\}_{\substack{Q\in\mathcal{D} \\ Q\subset Q_0 \\ s(Q)\leq \mathrm{const}_{Q_0} \\ |(c_Q)_N| \geq c\cdot s(Q)}} \in l_p.$$

Here, we use the shorthands

$$\mathrm{const}_{Q_0} = \frac{1}{2N^{\frac{1}{2}}} \min\{ \inf_{\substack{x\in Q_0 \\ v_0\neq v\in R}} |\langle x, v\rangle|, \inf_{\substack{x\in Q_0 \\ v_0\neq v\in R}} |\langle gx, v\rangle| \}, \quad c = \frac{2N^{\frac{1}{2}}}{|(v_0)_j|}.$$

Using Lemmas 2.4, 6.4 and 6.5, we write

$$\begin{aligned} & w(O_{v_0}Q)^{-1} \langle |f| \cdot B_j(\chi_{gO_{v_0}Q}), \chi_{O_{v_0}Q}\rangle_{L_2(\mathbb{R}^N, w(x)dx)} = \\ &= w(O_{v_0}Q)^{-1} \int_{O_{v_0}Q\times gO_{v_0}Q} |f|(x)|K_j|(x,y)w(x)w(y)dxdy \geq \\ &\overset{L.2.4}{\geq} c_k^{(1)} w(O_{v_0}Q)^{-1} \int_{O_{v_0}Q\times gO_{v_0}Q} |f|(x) \frac{|(x-y)_j|}{|x-y|} \cdot \frac{w(x)w(y)}{w(B(x,|x-y|))} dxdy \geq \\ &\overset{L.6.4(\mathrm{iii})}{\geq} c_k^{(2)} w(O_{v_0}Q)^{-1} \int_{O_{v_0}Q\times gO_{v_0}Q} |f|(x) \frac{|(x-y)_j|}{|x-y|} \cdot \frac{w^2(O_{v_0}c_Q)}{w(B(x,|x-y|))} dxdy \geq \\ &\overset{L.6.4(\mathrm{iv})}{\geq} c_k^{(3)} w(O_{v_0}Q)^{-1} \int_{O_{v_0}Q\times gO_{v_0}Q} |f|(x) \frac{|(x-y)_j|}{|(c_Q)_N|} \cdot \frac{w^2(O_{v_0}c_Q)}{w(B(x,|x-y|))} dxdy \geq \\ &\overset{L.6.4(\mathrm{vi})}{\geq} c_k^{(4)} m(Q)^{-1} \int_{O_{v_0}Q\times gO_{v_0}Q} |f|(x) \frac{|(x-y)_j|}{|(c_Q)_N|} \cdot \frac{w(O_{v_0}c_Q)}{w(B(x,|x-y|))} dxdy \geq \end{aligned}$$

$$\overset{L.6.4(v)}{\geq} c^{(5)}_{k,Q_0} m(Q)^{-1} \int_{O_{v_0}Q\times gO_{v_0}Q} |f|(x)|(x-y)_j| \cdot \frac{1}{|(c_Q)_N|^{N+1}} dxdy =$$

$$= c^{(5)}_{k,Q_0} \frac{1}{m(Q)\cdot |(c_Q)_N|^{N+1}} \int_{O_{v_0}Q} |f|(x)\Big(\int_{gO_{v_0}Q} |(x-y)_j| dy\Big) dx \geq$$

$$\overset{L.6.5}{\geq} c^{(6)}_{k,Q_0} \frac{1}{|(c_Q)_N|^{N}} \int_{O_{v_0}Q} |f|(x) dx.$$

As the condition $s(Q) \leq \mathrm{const}_{Q_0}$ excludes only finitely many terms, we can drop this condition. □

In case v_0 is orthogonal to e_j, we similarly obtain

$$\Big\{\frac{1}{|(c_Q)_N|^{N+1}} \int_{O_{v_0}Q} |f|(x)|(x-O_{v_0}c_Q)_j| dx\Big\}_{Q\in\mathcal{D}_{v_0,Q_0}} \in l_p.$$

Of course, in this case, we should omit $|(v_0)_j|$ from the definition of $\mathcal{D}_{v_0,Q_0}$.

7. Proof of Theorem 1.1 (ii)

Lemma 7.1. *Let Γ be a chamber and let $x_0 \in \Gamma$. We have*

$$\sup_{x\in B(x_0,r)\cap\Gamma} w(x) \leq \frac{c_{k,\Gamma}}{m(B(x_0,r)\cap\Gamma)} \int_{B(x_0,r)\cap\Gamma} w(x) dx.$$

Proof. If $x = x_0 + y$, $y \in B(0,r)$, then

$$|\langle x, v\rangle| \leq |\langle x_0, v\rangle| + r|\langle y, v\rangle| \leq |\langle x_0, v\rangle| + r\sqrt{2}.$$

Thus,

$$w(x) = \prod_{v\in R} (|\langle x_0, v\rangle| + |\langle y, v\rangle|)^{\epsilon\cdot k(v)} \leq \prod_{v\in R} (|\langle x_0, v\rangle| + r\sqrt{2})^{k(v)}.$$

In particular,

$$LHS \leq \prod_{v\in R} (|\langle x_0, v\rangle| + r\sqrt{2})^{k(v)}.$$

By convexity, $x_0 + \Gamma \subset \Gamma$. Hence,

$$x_0 + (B(0,r)\cap\Gamma) \subset B(x_0,r)\cap\Gamma.$$

So,

$$\int_{B(x_0,r)\cap\Gamma} w(x) dx \geq \int_{x_0+(B(0,r)\cap\Gamma)} w(x) dx = \int_{B(0,r)\cap\Gamma} w(x_0+y) dy.$$

If $y \in \Gamma$, then

$$|\langle x_0 + y, v\rangle| = |\langle x_0, v\rangle| + |\langle y, v\rangle|, \quad v \in R.$$

Thus,

$$w(x_0+y) = \prod_{v\in R} (|\langle x_0, v\rangle| + |\langle y, v\rangle|)^{k(v)} \geq$$

$$\geq \max\{\prod_{v\in R} |\langle x_0, v\rangle|^{k(v)}, \prod_{v\in R} |\langle y, v\rangle|^{k(v)}\}.$$

Hence,

$$\int_{B(x_0,r)\cap\Gamma} w(x) dx \geq \int_{B(0,r)\cap\Gamma} w(x_0+y) dy \geq$$

$$\geq \max\Big\{\int_{B(0,r)\cap\Gamma}\prod_{v\in R}|\langle x_0,v\rangle|^{k(v)}dy, \int_{B(0,r)\cap\Gamma}\prod_{v\in R}|\langle y,v\rangle|^{k(v)}dy\Big\}.$$

Obviously,

$$\int_{B(0,r)\cap\Gamma}\prod_{v\in R}|\langle x_0,v\rangle|^{k(v)}dy = r^N\prod_{v\in R}|\langle x_0,v\rangle|^{k(v)}\int_{B(0,1)\cap\Gamma}dy,$$

$$\int_{B(0,r)\cap\Gamma}\prod_{v\in R}|\langle y,v\rangle|^{k(v)}dy = r^{N+\sum_{v\in R}k(v)}\int_{B(0,1)\cap\Gamma}\prod_{v\in R}|\langle y,v\rangle|^{k(v)}dy.$$

Thus,

$$\int_{B(x_0,r)\cap\Gamma}w(x)dx \geq c^{(1)}_{k,\Gamma}\max\{r^N\prod_{v\in R}|\langle x_0,v\rangle|^{k(v)}, r^{N+\sum_{v\in R}k(v)}\}.$$

Combining this with the preceding paragraph, we complete the proof. □

Lemma 7.2. *Let Γ be a chamber and let $x_0\in\Gamma$. Suppose $1<t<1+\frac{1}{|R|\cdot\max_{v\in R}k(v)}$. For every $r>0$, we have*

$$\Big(\frac{\int_{B(x_0,r)\cap\Gamma}w^{1-t}(x)dx}{\int_{B(x_0,r)\cap\Gamma}w(x)dx}\Big)^{\frac{1}{t}} \leq c_{t,k,\Gamma}\frac{m(B(x_0,r)\cap\Gamma)}{\int_{B(x_0,r)\cap\Gamma}w(x)dx}.$$

Proof. If $v\in R$ is such that

$$2\sqrt{2}r\leq|\langle x_0,v\rangle|,$$

then

$$|\langle x,v\rangle|\geq|\langle x_0,v\rangle|-|\langle x-x_0,v\rangle|\geq|\langle x_0,v\rangle|-r\sqrt{2}\geq\frac{1}{2}|\langle x_0,v\rangle|,\quad x\in B(x_0,r).$$

In this case,

$$\int_{B(x_0,r)}|\langle x,v\rangle|^{(1-t)|R|k(v)}dx\leq 2^{(t-1)|R|k(v)}|\langle x_0,v\rangle|^{(1-t)|R|k(v)}\cdot\int_{B(x_0,r)}dx=$$

$$=2^{(t-1)|R|k(v)}|\langle x_0,v\rangle|^{(1-t)|R|k(v)}\cdot\mathrm{Vol}(\mathbb{B}^N)r^N.$$

If $v\in R$ is such that

$$|\langle x_0,v\rangle|\leq 2\sqrt{2}r,$$

then let $y_0=\langle x_0,v\rangle$ and $z_0=x_0-\frac{1}{2}y_0v_0$. Note that $|y_0|\leq 2\sqrt{2}r$. Using new variables $y=\langle x,v\rangle$ and $z=x-\frac{1}{2}yv$, we write

$$\int_{B(x_0,r)}|\langle x,v\rangle|^{(1-t)|R|k(v)}dx=\int_{|y-y_0|^2+|z-z_0|^2\leq r^2}|y|^{(1-t)|R|k(v)}dydz\leq$$

$$\leq\int_{|y-y_0|^2,|z-z_0|^2\leq r^2}|y|^{(1-t)|R|k(v)}dydz=$$

$$=r^{N-1}\mathrm{Vol}(\mathbb{B}^{N-1})\cdot\int_{|y-y_0|\leq r}|y|^{(1-t)|R|k(v)}dy\leq$$

$$=r^{N-1}\mathrm{Vol}(\mathbb{B}^{N-1})\cdot\int_{|y|\leq(1+2\sqrt{2})r}|y|^{(1-t)|R|k(v)}dy=$$

$$=r^{N+(1-t)|R|k(v)}\mathrm{Vol}(\mathbb{B}^{N-1})\cdot\int_{|y|\leq 1+2\sqrt{2}}|y|^{(1-t)|R|k(v)}dy.$$

Combining the estimates in the preceding paragraphs, we write

$$\int_{B(x_0,r)} |\langle x,v\rangle|^{(1-t)|R|k(v)}dx \le c_1 R^n \big(\max\{|\langle x_0,v\rangle|,r\}\big)^{(1-t)|R|k(v)}.$$

By Hölder inequality,

$$\int_{B(x_0,r)} w^{1-t}(x)dx \le \prod_{v\in R}\Big(\int_{B(x_0,r)} |\langle x,v\rangle|^{(1-t)|R|k(v)}dx\Big)^{\frac{1}{|R|}} \le$$

$$\le c_2 R^N \prod_{v\in R}\big(\max\{|\langle x_0,v\rangle|,r\}\big)^{(1-t)k(v)}.$$

The assertion follows easily. □

Lemma 7.3. *Let Γ be a chamber equipped with Euclidean distance. Measures dx and $w(x)dx$ and operator $T_{\Gamma,j} = M_{\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_\Gamma}$ satisfy on Γ the conditions in Theorem 1.4 in [8].*

Proof. **Step 1:** We claim that measures dx and $w(x)dx$ are A_∞-equivalent (in the sense of Proposition 1.2 in [8]). This verifies Assumption (i) in Theorem 1.4 in [8].

Let $x_0 \in \Gamma$. By Lemmas 7.1 and 7.2,

$$\Big(\frac{1}{m(B\cap\Gamma)}\int_{B\cap\Gamma} w^t(x)dx\Big)^{\frac{1}{t}} \le \frac{c_{k,\Gamma}}{m(B\cap\Gamma)}\int_{B\cap\Gamma} w(x)dx, \quad t>0.$$

$$\Big(\frac{\int_{B\cap\Gamma} w^{1-t}(x)dx}{\int_{B\cap\Gamma} w(x)dx}\Big)^{\frac{1}{t}} \le c_G \frac{m(B\cap\Gamma)}{\int_{B\cap\Gamma} w(x)dx}, \quad 1<t<1+\frac{1}{|R|\cdot\max_{v\in R}k(v)}.$$

The assertion follows now from Proposition 3.6 in [8].

Step 2: Assumption (ii) in Theorem 1.4 in [8] is obvious.

Step 3: Integral kernel K_j of the operator $T_{\Gamma,j}$ satisfies the estimates

$$|K_j(x,y)| \le \frac{c_G}{w(B(x,|x-y|)\cap\Gamma)}, \quad x,y\in\Gamma,$$

$$|K_j(x,y)-K_j(x',y)| \le \frac{|x-x'|}{|x-y|}\cdot\frac{c_G}{w(B(x,|x-y|))}, \quad |x-x'|\le\frac{1}{2}|x-y|, \quad x,x',y\in\Gamma,$$

$$|K_j(x,y)-K_j(x,y')| \le \frac{|y-y'|}{|x-y|}\cdot\frac{c_G}{w(B(x,|x-y|))}, \quad |y-y'|\le\frac{1}{2}|x-y|, \quad x,y,y'\in\Gamma.$$

In other words, K_j is a standard kernel for the measure $w(x)dx$.

This follows from Theorem 1.1 in [5].

Step 4: Kernel K_j is non-degenerate in the following sense: for every $x\in\Gamma$ and $r>0$, there exists $y\in\Gamma$ with $|x-y|=r$ and

$$|K_j(x,y)| \ge \frac{c_G}{w(B(x,|x-y|)\cap\Gamma)}.$$

Let

$$c_\Gamma = (\max_{z\in\Gamma,\ |z|=1}|z_j|)^{-1}.$$

Choose $z\in\Gamma$ such that $|z_j| = c_\Gamma^{-1}$ and $|z|=1$. Set $y = x+rz$. Since $x,z\in\Gamma$, it follows that $y\in\Gamma$. Clearly, $|(x-y)_j|\ge rc_\Gamma^{-1}$ and $|x-y|=r$. The claim follows from the proof of the Theorem 1.2 in [5].

Step 4: Lebesgue measure on Γ is, obviously, N-regular and possesses the $(1,1)$ Poincare inequality. □

Lemma 7.4. *If* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$, *then* $f \in \dot{W}^{\frac{N}{p},p}(\Gamma)$ *for every chamber* Γ. *Furthermore,*

$$\|f\|_{\dot{W}^{\frac{N}{p},p}(\Gamma)} \leq c_{R,k} \|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}.$$

Proof. This follows immediately from Lemma 7.3 and Theorem 1.4 in [8]. □

Let $U_g : L_2(\mathbb{R}^N, w(x)dx) \to L_2(\mathbb{R}^N, w(x)dx)$ be given by the formula $U_g\xi = \xi \circ g^{-1}$. Denote for brevity

$$A_{g,\Gamma,j} = M_{\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_{g\Gamma}} U_g.$$

Lemma 7.5. *If* $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$, *then*[3]

$$M_{(f-f\circ g)\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_{g\Gamma}} \in \mathcal{L}_p(L_2(\Gamma, w(x)dx))$$

for every $g \in G$ *and for every chamber* Γ. *Furthermore,*

$$\|M_{(f-f\circ g)\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_{g\Gamma}}\|_{\mathcal{L}_p(L_2(\Gamma, w(x)dx))} \leq c_{R,k} \|[R_{\mathrm{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}.$$

Proof. By assumption,

$$M_{\chi_\Gamma}[R_{\mathrm{Dunkl},j}, M_f] M_{\chi_{g\Gamma}} U_g \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx)).$$

We, therefore, have

$$A_{g,\Gamma,j} M_{f\circ g} - M_f A_{g,\Gamma,j} \in \mathcal{L}_p(L_2(\Gamma, w(x)dx)). \tag{7.1}$$

Let us denote by $K_{g,\Gamma,j}$ the integral kernel of the operator $A_{g,\Gamma,j}$. By Lemma 2.2,

$$|K_{g,\Gamma,j}|(x,y) \leq \frac{c_G}{w(B(x, |x-y|))}, \quad x, y \in \Gamma.$$

We know from Lemma 7.4 that $f \in \dot{W}^{\frac{N}{p},p}(\Gamma)$. We now infer from the Proposition 4.5 that

$$A_{g,\Gamma,j} M_{f\circ g} - M_{f\circ g} A_{g,\Gamma,j} \in \mathcal{L}_p(L_2(\Gamma, w(x)dx)). \tag{7.2}$$

Combining (7.1) and (7.2), we write

$$M_{f-f\circ g} A_{g,\Gamma,j} \in \mathcal{L}_p(L_2(\Gamma, w(x)dx)).$$

Multiplying by U_g^{-1} on the right, we complete the proof. □

[Zhijie may move the following definition to somewhere when writing the introduction]

Lemma 7.6. *For any* $1 \leq p < \infty$ *and* $b \in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$, *we have*

$$\|b\|_{B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)} \lesssim \|b\|_{B^p(\mathbb{R}^N, dw)} + \|b\|_{B^p_{\mathrm{Non-standard}}(\mathbb{R}^N)}, \tag{7.3}$$

where we write

$$\|b\|_{B^p(\mathbb{R}^N, dw)} := \left(\int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \frac{|b(x) - b(y)|^p}{w(B(x, \|x-y\|))^2} dw(x) dw(y) \right)^{1/p}$$

[3] this footnote is temporary, for the authors' use only. the combination of assertions of Lemmas 7.4 and 7.5 yield $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$.

and

$$\|b\|_{B^p_{\mathrm{Non-standard}}(\mathbb{R}^N)} := \Bigg(\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}} \sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p}\times$$
$$\times \fint_{B(c_Q,A_2 2^{-k+1})} \fint_{B(\widetilde{\sigma}(\sigma(c_Q)),A_2 2^{-k+1})} |b(x)-b(y)|^p dw(x)dw(y)\Bigg)^{1/p}.$$

Proof. Let Γ be a fixed Weyl chamber. By decomposing

$$\mathbb{R}^N = \bigcup_{\sigma\in G}\sigma(\Gamma),$$

we write

$$\begin{aligned}
\|b\|^p_{B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)} &= \sum_{\sigma,\widetilde{\sigma}\in G}\int_{\sigma(\Gamma)}\int_{\widetilde{\sigma}(\Gamma)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p \frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&= \sum_{\sigma\in G}\int_{\sigma(\Gamma)}\int_{\sigma(\Gamma)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p \frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&\quad+\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\int_{\sigma(\Gamma)}\int_{\widetilde{\sigma}(\Gamma)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p \frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&= \sum_{\sigma\in G}\int_{\sigma(\Gamma)}\int_{\sigma(\Gamma)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p \frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&\quad+\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\int_{\sigma(\Gamma)}\int_{D(y)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p \frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&\quad+\sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}}\int_{\sigma(\Gamma)}\int_{E(y)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p \frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y),
\end{aligned}\tag{7.4}$$

where we denote by

$$D(y) := \left\{x\in\widetilde{\sigma}(\Gamma):\ \|x-y\|<2^{10}d(x,y)\right\},$$

$$E(y) := \left\{x\in\widetilde{\sigma}(\Gamma):\ \|x-y\|\geq 2^{10}d(x,y)\right\}.$$

It follows from (2.4) that $d(x,y)=\|x-\widetilde{\sigma}(\sigma(y))\|$ if $x\in\widetilde{\sigma}(\Gamma)$ and $y\in\sigma(\Gamma)$ and that $d(x,y)\leq\|x-y\|$ for all $x,y\in\mathbb{R}^N$. In particular, to estimate the first term on the right-hand side of (7.4), we note that for each $\sigma\in G$ and for every $x,y\in\sigma(\Gamma)$, $d(x,y)=\|x-y\|$. To estimate the second term, we apply the doubling inequality (2.2) to replace $d(x,y)$ by $\|x-y\|$. Consequently, the first two terms are dominated by $C\|b\|_{B^p(\mathbb{R}^N,dw)}$, where we used the facts that G is a finite group and w is a doubling measure. Thus, it remains to estimate the third term.

Let $\mathbb{D}_\Gamma := \bigcup_{k\in\mathbb{Z}}\mathbb{D}_{\Gamma,k}$ be a system of dyadic cubes over Γ, which is constructed in Corollary **??**. It follows that $\sigma(\mathbb{D}_\Gamma) := \bigcup_{k\in\mathbb{Z}}\sigma(\mathbb{D}_{\Gamma,k})$ is a system of dyadic cubes

over $\sigma(\Gamma)$. Recall that A_2 is the constant given in Definition **??**. We have

$$\begin{aligned}
&\int_{\sigma(\Gamma)}\int_{E(y)}\left(\frac{d(x,y)}{\|x-y\|}\right)^p\frac{|b(x)-b(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\\
&=\int_{\sigma(\Gamma)}\int_{E(y)}\left(\frac{\|x-\widetilde{\sigma}(\sigma(y))\|}{\|x-y\|}\right)^p\frac{|b(x)-b(y)|^p}{w(B(x,\|x-\widetilde{\sigma}(\sigma(y))\|))^2}dw(x)dw(y)\\
&\le\sum_{k\in\mathbb{Z}}\int_{\sigma(\Gamma)}\int_{E_k(y)}\left(\frac{A_2 2^{-k}}{\|x-y\|}\right)^p\frac{|b(x)-b(y)|^p}{w(B(x,A_2 2^{-k-1}))^2}dw(x)dw(y)\\
&=\sum_{k\in\mathbb{Z}}\sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})}\int_Q\int_{E_k(y)}\left(\frac{A_2 2^{-k}}{\|x-y\|}\right)^p\frac{|b(x)-b(y)|^p}{w(B(x,A_2 2^{-k-1}))^2}dw(x)dw(y),
\end{aligned}\tag{7.5}$$

where

$$E_k(y):=\left\{x\in E(y):\ A_2 2^{-k-1}\le\|x-\widetilde{\sigma}(\sigma(y))\|\le A_2 2^{-k}\right\}.$$

It follows from the doubling inequality (2.2) and the G-invariance of the Dunkl measure that for every $y\in Q$ and $x\in E_k(y)\subset B(\widetilde{\sigma}(\sigma(c_Q)),A_2 2^{-k+1})$ with $Q\in\sigma(\mathbb{D}_{\Gamma,k+10})$,

$$w(B(x,A_2 2^{-k-1}))\sim w(B(\widetilde{\sigma}(\sigma(c_Q)),2^{-k}))\sim w(B(c_Q,2^{-k}))\sim w(Q).$$

Substituting this inequality into (7.5), we deduce that

$$\begin{aligned}
&\text{RHS of (7.5)}\\
&\lesssim\sum_{k\in\mathbb{Z}}\sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})}\frac{2^{-kp}}{w(B(c_Q,2^{-k}))}\fint_Q\int_{E_k(y)}\left(\frac{|b(x)-b(y)|}{\|x-y\|}\right)^p dw(x)dw(y)\\
&\lesssim\sum_{k\in\mathbb{Z}}\sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})}\sum_{\ell\in\mathbb{Z}}\frac{2^{-kp}}{w(B(c_Q,2^{-k}))}\fint_Q\int_{E_{k,\ell}(y)}\left(\frac{|b(x)-b(y)|}{\|x-y\|}\right)^p dw(x)dw(y)\\
&\lesssim\sum_{k\in\mathbb{Z}}\sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})}\sum_{\ell\in\mathbb{Z}}\frac{2^{(\ell-k)p}}{w(B(c_Q,2^{-k}))}\fint_Q\int_{E_{k,\ell}(y)}|b(x)-b(y)|^p dw(x)dw(y),
\end{aligned}\tag{7.6}$$

where

$$E_{k,\ell}(y):=\left\{x\in E_k(y):\ A_2 2^{-\ell-1}\le\|x-y\|\le A_2 2^{-\ell}\right\}.$$

Note that for every $y\in\sigma(\Gamma)$ and $x\in E_{k,\ell}(y)$,

$$A_2 2^{-k-1}\le\|x-\widetilde{\sigma}(\sigma(y))\|=d(x,y)\le 2^{-10}\|x-y\|\le A_2 2^{-\ell-10}.$$

This implies that the summands on the right-hand side of (7.6) vanish provided $\ell>k-9$. To continue, for every $Q\in\mathbb{D}_\Gamma$, we denote by

$$U_k(Q):=B(c_Q,A_2 2^{-k+1})\backslash B(c_Q,A_2 2^{-k-2}).$$

Note that the reflection σ is linear and preserves the Euclidean norm. Therefore, if $y\in Q\in\sigma(\mathbb{D}_{\Gamma,k+10})$ and $x\in E_{k,\ell}(y)$, then

$$\|x-\widetilde{\sigma}(\sigma(c_Q))\|\le\|x-\widetilde{\sigma}(\sigma(y))\|+\|\widetilde{\sigma}(\sigma(y))-\widetilde{\sigma}(\sigma(c_Q))\|\le A_2 2^{-k}+A_2 2^{-(k+10)}\le A_2 2^{-(k+1)},$$

$$\|x-\widetilde{\sigma}(\sigma(c_Q))\|\ge\|x-\widetilde{\sigma}(\sigma(y))\|-\|\widetilde{\sigma}(\sigma(y))-\widetilde{\sigma}(\sigma(c_Q))\|\ge A_2 2^{-k-1}-A_2 2^{-(k+10)}\ge A_2 2^{-(k-2)},$$

$$\|x-c_Q\|\le\|x-y\|+\|y-c_Q\|\le A_2 2^{-\ell}+A_2 2^{-(k+10)}\le A_2 2^{-\ell+1},$$

$$\|x-c_Q\|\ge\|x-y\|-\|y-c_Q\|\ge A_2 2^{-\ell-1}-A_2 2^{-(k+10)}\ge A_2 2^{-\ell-2}.$$

In other words, we have $x \in U_k(\widetilde{\sigma}(\sigma(Q))) \cap U_\ell(Q)$. Combining these facts, we see that

$$\begin{aligned}
&\text{RHS of (7.6)}\\
&= \sum_{k\in\mathbb{Z}} \sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})} \sum_{\ell\le k-9} \frac{2^{(\ell-k)p}}{w(B(c_Q,2^{-k}))} \fint_Q \int_{E_{k,\ell}(y)} |b(x)-b(y)|^p dw(x)dw(y)\\
&\lesssim \sum_{k\in\mathbb{Z}} \sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})} \sum_{\ell\le k-9} \frac{2^{(\ell-k)p}}{w(B(c_Q,2^{-k}))} \fint_Q \int_{E_{k,\ell}} |b(x)-b(y)|^p dw(x)dw(y),
\end{aligned} \tag{7.7}$$

where

$$E_{k,\ell} := \widetilde{\sigma}(\Gamma) \cap U_k(\widetilde{\sigma}(\sigma(Q))) \cap U_\ell(Q).$$

We *claim* that if $\ell \ge -\log_2(A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|) + 3$, then

$$B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1}) \cap B(c_Q, A_2 2^{-\ell+1}) = \varnothing. \tag{7.8}$$

In particular, we have $U_k(\widetilde{\sigma}(\sigma(Q))) \cap U_\ell(Q) = \varnothing$. If this is not the case, then there exists an element $x_0 \in B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1}) \cap B(c_Q, A_2 2^{-\ell+1})$. Equivalently, we have $\|x_0 - \widetilde{\sigma}(\sigma(c_Q))\| \le A_2 2^{-k+1}$ and $\|x_0 - c_Q\| \le A_2 2^{-\ell+1}$. Together with the inequality $\ell \le k-9$, we see that

$$\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\| \le \|x_0 - \widetilde{\sigma}(\sigma(c_Q))\| + \|x_0 - c_Q\| \le A_2 2^{-k+1} + A_2 2^{-\ell+1} \le A_2 2^{-\ell+2}.$$

It follows from the assumption $\ell \ge -\log_2(A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|) + 3$ that

$$A_2 2^{-\ell+2} \le \frac{1}{2}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|.$$

Combining with the above two inequalities, we deduce that $c_Q = \widetilde{\sigma}(\sigma(c_Q)) \in \sigma(\Gamma) \cap \widetilde{\sigma}(\Gamma)$, which is a contradiction since each c_Q does not lie in the boundaries of the Weyl chambers. This ends the proof of (7.8).

Next, we *claim* that if $\ell \le \min\left\{k-4, -\log_2(A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|) - 3\right\}$, then

$$B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1}) \subset B(c_Q, A_2 2^{-\ell-2}). \tag{7.9}$$

In particular, we have $U_k(\widetilde{\sigma}(\sigma(Q))) \cap U_\ell(Q) = \varnothing$. Indeed, we take x to be an arbitrary element of $B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1})$, then

$$\|x - c_Q\| \le \|x - \widetilde{\sigma}(\sigma(c_Q))\| + \|c_Q - \widetilde{\sigma}(\sigma(c_Q))\| \le A_2 2^{-k+1} + \|c_Q - \widetilde{\sigma}(\sigma(c_Q))\| \le A_2 2^{-\ell-2},$$

where the last inequality follows from the assumption. This ends the proof of (7.9).

Denote by

$$\ell_{k,Q} := \min\left\{k-4, -\log_2(A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|) - 3\right\},$$

$$L_{k,Q} := \min\left\{k-9, -\log_2(A_2^{-1}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|) + 3\right\}.$$

Combining (7.8) with (7.9), we see that

RHS of (7.7)

$$= \sum_{k\in\mathbb{Z}} \sum_{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})} \sum_{\ell_{k,Q}<\ell\le L_{k,Q}} \frac{2^{(\ell-k)p}}{w(B(c_Q,2^{-k}))} \fint_Q \int_{E_{k,\ell}} |b(x)-b(y)|^p dw(x)dw(y)$$

$$= \sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|}} \sum_{-3<\ell+\log_2(A_2^{-1}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|)\le 4} \frac{2^{(\ell-k)p}}{w(B(c_Q,2^{-k}))}\times$$

$$\times \fint_Q \int_{E_{k,\ell}} |b(x)-b(y)|^p dw(x)dw(y). \tag{7.10}$$

Next, using the trivial inclusions $E_{k,\ell}\subset B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1})$ and $Q\subset B(c_Q, A_2 2^{-k+1})$, the doubling inequality (2.2) and the G-invariance of the Dunkl measure, we conclude that the right-hand side of (7.10) is dominated by

$$C\sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p} \fint_{B(c_Q,A_2 2^{-k+1})} \fint_{B(\widetilde{\sigma}(\sigma(c_Q)),A_2 2^{-k+1})} |b(x)-b(y)|^p dw(x)dw(y).$$

This ends the proof of Lemma 7.6. □

With the collection $\{\mathbb{D}^\nu\}_{\nu=1}^{2^N}$ of dyadic cubes systems given in Lemma **??**, we define the conditional expectation of a locally integrable function f on $\mathbb{R}^N$ by the expression

$$E_k^\nu(f)(x) := \sum_{Q\in\mathbb{D}_k^\nu} (f)_Q \chi_Q(x),\ x\in\mathbb{R}^N.$$

Moreover, we define the martingale difference of f by

$$D_k^\nu(f) := E_{k+1}^\nu(f) - E_k^\nu(f).$$

Definition 7.7. *For any $1\le p<\infty$ and $f\in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$, we define the martingale Besov (semi)norm of f by*

$$\|f\|_{MB^p(\mathbb{R}^N,dw)} := \sum_{\nu=1}^{2^N} \Big(\sum_{k\in\mathbb{Z}} \int_{\mathbb{R}^N} \frac{|D_k^\nu f(x)|^p}{w(B(x,2^{-k}))} dw(x)\Big)^{1/p}.$$

Lemma 7.8. *Let T be a bounded linear operator on a Hilbert space H. For any $\xi,\eta\in H$, we have*

$$\langle T\xi,\eta\rangle \ = \ \mathrm{Tr}\big((\xi\otimes\eta)T\big),$$

where $\xi\otimes\eta$ denotes the rank-one operator defined by $(\xi\otimes\eta)(x) := \langle x,\eta\rangle\xi$ for $x\in H$.

Proof. For any $x\in H$, we have

$$(\xi\otimes\eta)T(x) = (\xi\otimes\eta)(Tx) = \langle Tx,\eta\rangle\xi = \langle x,T^*\eta\rangle\xi = (\xi\otimes T^*\eta)(x).$$

Hence,

$$(\xi\otimes\eta)T = \xi\otimes T^*\eta.$$

Since the trace of a rank-one operator $u\otimes v$ is given by $\mathrm{Tr}(u\otimes v) = \langle u,v\rangle$, we obtain

$$\mathrm{Tr}((\xi\otimes\eta)T) = \mathrm{Tr}(\xi\otimes T^*\eta) = \langle \xi, T^*\eta\rangle.$$

Finally, $\langle \xi, T^*\eta\rangle = \langle T\xi,\eta\rangle$, which completes the proof. □

Lemma 7.9. *Suppose that $N < p < \infty$ and $b \in L^1_{\mathrm{loc}}(\mathbb{R}^N, dw)$. Assume that $[b, R_{\mathrm{Dunkl},j}] \in S^p$ for some $j \in \{1, 2, \dots, N\}$, then we have*

$$\|b\|_{B^p_{\mathrm{Non-standard}}(\mathbb{R}^N)} \lesssim \|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}.$$

Proof. Let $k_0 := \lfloor \log_2(24A_2) \rfloor$ be an integer such that $\frac{2^{-(k-k_0)-2}}{3} \le A_2 2^{-k+1} < \frac{2^{-(k-k_0)-1}}{3}$. By Lemma **??**, for each $k \in \mathbb{Z}$ and $Q \in \sigma(\mathbb{D}_{\Gamma,k+10})$, there exist cubes $W_Q, P_Q \in \bigcup_{\nu=1}^{2^N} \mathbb{D}^\nu_{k-k_0}$ such that

$$B(c_Q, A_2 2^{-k+1}) \subset W_Q \subset B(c_Q, 12\sqrt{N} A_2 2^{-k+1}), \tag{7.11}$$

$$B(\widetilde{\sigma}(\sigma(c_Q)), A_2 2^{-k+1}) \subset P_Q \subset B(\widetilde{\sigma}(\sigma(c_Q)), 12\sqrt{N} A_2 2^{-k+1}). \tag{7.12}$$

Together with the doubling inequality (2.2), we deduce that

$$\begin{aligned}
&\|b\|^p_{B^p_{\mathrm{Non-standard}}(\mathbb{R}^N)} \\
&\lesssim \sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}} \sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le \|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p} \fint_{W_Q}\fint_{P_Q} |b(x)-b(y)|^p dw(x)dw(y) \\
&= \sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}} \sum_{\nu,\tau=1}^{2^N} \sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le \|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ P_Q\in\mathbb{D}^\nu_{k-k_0}, W_Q\in\mathbb{D}^\tau_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p} \fint_{W_Q}\fint_{P_Q} |b(x)-b(y)|^p dw(x)dw(y) \\
&\lesssim \sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}} \sum_{\tau=1}^{2^N} \sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le \|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ W_Q\in\mathbb{D}^\tau_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p} \fint_{W_Q} |b(x)-(b)_{W_Q}|^p dw(x) \\
&\quad + \sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}} \sum_{\nu=1}^{2^N} \sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le \|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ P_Q\in\mathbb{D}^\nu_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p} \fint_{P_Q} |b(x)-(b)_{P_Q}|^p dw(x) \\
&\quad + \sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}} \sum_{\nu,\tau=1}^{2^N} \sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le \|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|\\ P_Q\in\mathbb{D}^\nu_{k-k_0}, W_Q\in\mathbb{D}^\tau_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p} |(b)_{P_Q}-(b)_{W_Q}|^p \\
&=: \sum_{\substack{\sigma,\widetilde{\sigma}\in G\\ \sigma\neq\widetilde{\sigma}}} (\mathrm{I}^p_{\sigma,\widetilde{\sigma}} + \mathrm{II}^p_{\sigma,\widetilde{\sigma}} + \mathrm{III}^p_{\sigma,\widetilde{\sigma}}).
\end{aligned}$$

Since $Q \subset W_Q$ for all $Q \in \sigma(\mathbb{D}_{\Gamma,k+10})$, we see that for each $\widetilde{Q} \in \mathbb{D}^\tau_{k-k_0}$, there are at most finite (an absolute constant) $Q \in \sigma(\mathbb{D}_{\Gamma,k+10})$ such that $W_Q = \widetilde{Q}$. This implies

that

$$\begin{aligned}\mathrm{I}^p_{\sigma,\tilde\sigma} &\lesssim \sum_{\tau=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ W_Q\in\mathbb{D}^\tau_{k-k_0}}} \fint_{W_Q}|b(x)-(b)_{W_Q}|^p dw(x)\\ &\lesssim \sum_{\tau=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\tau_{k-k_0}} \fint_{Q}|b(x)-(b)_{Q}|^p dw(x)\\ &\lesssim \|b\|^p_{B^p(\mathbb{R}^N,dw)}, \end{aligned}\tag{7.13}$$

where in the last inequality we used the fact that oscillation space $\mathrm{OSC}^{p,p}(\mathbb{R}^N,dw)$ is equivalent to Besov space $B^p(\mathbb{R}^N,dw)$ (see e.g., [9, Proposition 17.5]). Similarly, the second term $\mathrm{II}_{\sigma,\tilde\sigma}$ is also bounded by $\|b\|_{B^p(\mathbb{R}^N,dw)}$. Together with Lemma **??**, we deduce that

$$\mathrm{I}_{\sigma,\tilde\sigma}+\mathrm{II}_{\sigma,\tilde\sigma}\lesssim \|[b,R_{\mathrm{Dunkl},j}]\|_{S^p}.$$

Now we turn to the term $\mathrm{III}_{\sigma,\tilde\sigma}$. For each $Q\in\bigcup_{\nu=1}^{2^N}\mathbb{D}^\nu_{k-k_0}$ and $k\in\mathbb{N}$, denote by $Q^{[k]}$ the kth strict dyadic ancestor. For convenience, we also set $Q^{[0]}:=Q$. Pick $\alpha(Q)\in\mathbb{Z}$ such that $2^{\alpha(Q)}\le A_2^{-1}\|c_Q-\widetilde\sigma(\sigma(c_Q))\|<2^{\alpha(Q)+1}$. Moreover, we let

$$s_Q:=\log_2(2^{k+10+\alpha(Q)})=k+10+\alpha(Q).$$

If $A_2 2^{-k+1}\le\|c_Q-\widetilde\sigma(\sigma(c_Q))\|$, then s_Q is a positive integer larger than 10. We have

$$\begin{aligned}\mathrm{III}^p_{\sigma,\tilde\sigma} &\lesssim \sum_{\nu=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le\|c_Q-\widetilde\sigma(\sigma(c_Q))\|\\ P_Q\in\mathbb{D}^\nu_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde\sigma(\sigma(c_Q))\|^{-p}|(b)_{P_Q}-(b)_{P_Q^{[s_Q]}}|^p\\ &\quad+\sum_{\tau=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le\|c_Q-\widetilde\sigma(\sigma(c_Q))\|\\ W_Q\in\mathbb{D}^\tau_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde\sigma(\sigma(c_Q))\|^{-p}|(b)_{W_Q}-(b)_{W_Q^{[s_Q]}}|^p\\ &\quad+\sum_{\nu,\tau=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10})\\ A_2 2^{-k+1}\le\|c_Q-\widetilde\sigma(\sigma(c_Q))\|\\ P_Q\in\mathbb{D}^\nu_{k-k_0},W_Q\in\mathbb{D}^\tau_{k-k_0}}} 2^{-kp}\|c_Q-\widetilde\sigma(\sigma(c_Q))\|^{-p}|(b)_{P_Q^{[s_Q]}}-(b)_{W_Q^{[s_Q]}}|^p\\ &=:\mathrm{III}^p_{\sigma,\tilde\sigma,1}+\mathrm{III}^p_{\sigma,\tilde\sigma,2}+\mathrm{III}^p_{\sigma,\tilde\sigma,3}.\end{aligned}$$

By Lemma **??**, for each $m\in\mathbb{N}$ and for each $Q\in\sigma(\mathbb{D}_\Gamma)$ with $P_Q\in\mathbb{D}^\nu_{k-k_0}$, there is a w-measurable function $H_{Q,m}$ on $\mathbb{R}^N$ (alternatively, $H_{Q,m}$ can be taken to be the Haar Basis element $h^1_{P_Q^{[m]}}$ (up to an absolute constant) if we enumerate $Ch(P_Q^{[m]})=\{R_1,R_2,\dots,R_{2^N}\}$ so that $R_1=P_Q^{[m-1]}$) satisfying

(1) $|H_{Q,m}(x)|\le w(P_Q^{[m]})^{-1/2}\chi_{P_Q^{[m]}}(x)$;

(2) $\int_{P_Q^{[m]}}H_{Q,m}(x)dw(x)=0$;

(3) $|(b)_{P_Q^{[m-1]}} - (b)_{P_Q^{[m]}}| \approx w(P_Q^{[m]})^{-1/2}|\langle b, H_{Q,m}\rangle_{L^2(\mathbb{R}^N,dw)}|$.

For simplicity, we denote by

$$\Lambda^\nu := \left\{(k,Q,m) : k\in\mathbb{Z}, Q\in\sigma(\mathbb{D}_{\Gamma,k+10}), A_2 2^{-k+1} \le \|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|, P_Q\in\mathbb{D}^\nu_{k-k_0}, 1\le m\le s_Q\right\}.$$

By Minkowski's inequality,

$$\begin{aligned}\mathrm{III}_{\sigma,\tilde{\sigma},1} &\lesssim \sum_{\nu=1}^{2^N}\left(\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p}|(b)_{P_Q^{[m-1]}} - (b)_{P_Q^{[m]}}|^p\right)^{1/p}\\ &\lesssim \sum_{\nu=1}^{2^N}\left(\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p} w(P_Q^{[m]})^{-p/2}|\langle b, H_{Q,m}\rangle_{L^2(\mathbb{R}^N,dw)}|^p\right)^{1/p}.\end{aligned} \tag{7.14}$$

By Lemma **??**, there exist NWO sequences $\{e^{(s)}_{P_Q^{[m]}}\}$ and $\{f^{(s)}_{P_Q^{[m]}}\}$, $s=1,2,3,4$, such that

$$w(P_Q^{[m]})^{-1/2}|\langle b, H_{Q,m}\rangle_{L^2(\mathbb{R}^N,dw)}| \lesssim \sum_{s=1}^4 \left|\left\langle [b, R_{\mathrm{Dunkl},j}]e^{(s)}_{P_Q^{[m]}}, f^{(s)}_{P_Q^{[m]}}\right\rangle_{L^2(\mathbb{R}^N,dw)}\right|.$$

Thus, the right-hand side of (7.14) is dominated by

$$\sum_{\nu=1}^{2^N}\sum_{s=1}^4\left(\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-kp}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-p}\left|\left\langle [b, R_{\mathrm{Dunkl},j}]e^{(s)}_{P_Q^{[m]}}, f^{(s)}_{P_Q^{[m]}}\right\rangle_{L^2(\mathbb{R}^N,dw)}\right|^p\right)^{1/p}.$$

Now taking $\{\varphi_{k,Q,m}\}_{(k,Q,m)\in\Lambda^\nu}$ with unit $\ell^{p'}$ norm, we apply the duality argument to estimate

$$\begin{aligned}&\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\left\langle [b, R_{\mathrm{Dunkl},j}]e^{(s)}_{P_Q^{[m]}}, f^{(s)}_{P_Q^{[m]}}\right\rangle_{L^2(\mathbb{R}^N,dw)}\varphi_{k,Q,m}\\ &= \sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\mathrm{Tr}\Big(\big(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}}\big)[b, R_{\mathrm{Dunkl},j}]\Big)\varphi_{k,Q,m}\\ &= \mathrm{Tr}\left(\Big(\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}})\Big)\cdot[b, R_{\mathrm{Dunkl},j}]\right)\\ &\le \left\|\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}})\right\|_{S^{p'}}\cdot\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p},\end{aligned} \tag{7.15}$$

where the first equality follows from Lemma 7.8. It remains to show that

$$\sup_{\|\{\varphi_{k,Q,m}\}_{(k,Q,m)\in\Lambda^\nu}\|_{\ell^{p'}}=1}\left\|\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}})\right\|_{S^{p'}}\lesssim 1. \tag{7.16}$$

To show (7.16), we first write

$$\begin{aligned}&\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}})\\ &=\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{Q\in\Theta^\nu_{m,k}} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}}),\end{aligned}\tag{7.17}$$

where

$$\Theta^\nu_{m,k}:=\left\{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}):A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|,\ \alpha(Q)\geq -k+m-10,\ P_Q\in\mathbb{D}^\nu_{k-k_0}\right\}.$$

Note that for any $Q\in\mathbb{D}^\nu_{k-k_0-m}$, there are at most $c2^{mN}$ cubes $W\in\sigma(\mathbb{D}_{\Gamma,k+10})$ such that $P_W^{[m]}=Q$, where c is an absolute constant. Denote these cubes by $Q^1_{[m]}$, $Q^2_{[m]}$, …, $Q^\iota_{[m]}$ with $\iota\leq c2^{mN}$. Let $\delta_{Q,m,\ell}:=1$ if $P^{[m]}_{Q^\ell_{[m]}}=Q$ and $\delta_{Q,m,\ell}:=0$ otherwise. Let

$$\Xi^\nu_{m,k,\ell}:=\left\{Q\in\mathbb{D}^\nu_{k-k_0-m}:A_2 2^{-k+1}\leq\|c_{Q^\ell_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|,\ \alpha(Q^\ell_{[m]})\geq -k+m-10\right\}.$$

We have

RHS of (7.17)

$$\begin{aligned}&=\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{\ell=1}^{c2^{mN}}\sum_{Q\in\Xi^\nu_{m,k,\ell}} 2^{-k}\|c_{Q^\ell_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|^{-1}\varphi_{k,Q^\ell_{[m]},m}\delta_{Q,m,\ell}(e^{(s)}_Q\otimes f^{(s)}_Q)\\ &=2^{-k_0}\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{\ell=1}^{c2^{mN}}\sum_{Q\in\Xi^\nu_{m,k+k_0+m,\ell}} 2^{-(k+m)}\|c_{Q^\ell_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|^{-1}\varphi_{k+k_0+m,Q^\ell_{[m]},m}\delta_{Q,m,\ell}(e^{(s)}_Q\otimes f^{(s)}_Q)\\ &=\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\nu_k}\lambda_Q(e^{(s)}_Q\otimes f^{(s)}_Q),\end{aligned}\tag{7.18}$$

where for any $Q\in\mathbb{D}^\nu_k$,

$$\begin{aligned}\lambda_Q:=2^{-k_0}\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}} &2^{-(k+m)}\|c_{Q^\ell_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|^{-1}\varphi_{k+k_0+m,Q^\ell_{[m]},m}\delta_{Q,m,\ell}\times\\ &\times\chi_{[A_2 2^{-(k+k_0+m-1)},\infty)}(\|c_{Q^\ell_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|)\chi_{[-(k+k_0-10),\infty)}(\alpha(Q^\ell_{[m]})).\end{aligned}$$

It follows from equalities (7.17), (7.18) and Lemma 2.6 that

$$\begin{aligned}&\left\|\sum_{(k,Q,m)\in\Lambda^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\varphi_{k,Q,m}(e^{(s)}_{P_Q^{[m]}}\otimes f^{(s)}_{P_Q^{[m]}})\right\|_{S^{p'}}\\ &=\left\|\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\nu_k}\lambda_Q(e^{(s)}_Q\otimes f^{(s)}_Q)\right\|_{S^{p'}}\\ &\lesssim\left(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\nu_k}|\lambda_Q|^{p'}\right)^{\frac{1}{p'}}.\end{aligned}\tag{7.19}$$

By Hölder's inequality and the definition of λ_Q,

$$\begin{aligned}
&\left(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}_k^\nu}|\lambda_Q|^{p'}\right)^{\frac{1}{p'}}\\
&\lesssim\left(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}_k^\nu}\left(\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}}2^{-(k+m)p}\|c_{Q^\ell_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|^{-p}\delta_{Q,m,\ell}\chi_{[-(k+k_0-10),\infty)}(\alpha(Q^\ell_{[m]}))\right)^{\frac{p'}{p}}\times\right.\\
&\qquad\times\left(\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}}\left|\varphi_{k+k_0+m,Q^\ell_{[m]},m}\delta_{Q,m,\ell}\chi_{[A_2 2^{-(k+k_0+m-1)},\infty)}(\|c_{Q^\ell_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|)\chi_{[-(k+k_0-10),\infty)}(\alpha(Q^\ell_{[m]}))\right|\right.
\end{aligned}$$

Since $p>N$, we have

$$\begin{aligned}
&\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}}2^{-(k+m)p}\|c_{Q^\ell_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|^{-p}\delta_{Q,m,\ell}\chi_{[-(k+k_0-10),\infty)}(\alpha(Q^\ell_{[m]}))\\
&\lesssim\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}}2^{-(k+m)p}\cdot 2^{kp}=C\sum_{m\geq 1}2^{-m(p-N)}\lesssim 1.
\end{aligned}$$

Combining this with the same reindexing and counting argument used to derive (7.18), we obtain

$$\begin{aligned}
&\left(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}_k^\nu}|\lambda_Q|^{p'}\right)^{\frac{1}{p'}}\\
&\lesssim\left(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}_k^\nu}\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}}\left|\varphi_{k+k_0+m,Q^\ell_{[m]},m}\times\right.\right.\\
&\qquad\left.\left.\times\,\delta_{Q,m,\ell}\chi_{[A_2 2^{-(k+k_0+m-1)},\infty)}(\|c_{Q^\ell_{[m]}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{[m]}}))\|)\chi_{[-(k+k_0-10),\infty)}(\alpha(Q^\ell_{[m]}))\right|^{p'}\right)^{\frac{1}{p'}}\\
&=\left(\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{\ell=1}^{c2^{mN}}\sum_{Q\in\Xi^\nu_{m,k+k_0+m,\ell}}\left|\varphi_{k+k_0+m,Q^\ell_{[m]},m}\right|^{p'}\right)^{\frac{1}{p'}}\\
&=\left(\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{\ell=1}^{c2^{mN}}\sum_{Q\in\Xi^\nu_{m,k,\ell}}\left|\varphi_{k,Q^\ell_{[m]},m}\right|^{p'}\right)^{\frac{1}{p'}}\\
&=\left(\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{Q\in\Theta^\nu_{m,k}}|\varphi_{k,Q,m}|^{p'}\right)^{\frac{1}{p'}}\\
&=\left(\sum_{(k,Q,m)\in\Lambda^\nu}|\varphi_{k,Q,m}|^{p'}\right)^{\frac{1}{p'}}\\
&=1.
\end{aligned}\tag{7.20}$$

Substituting this inequality into (7.19), we obtain the desired inequality (7.16). Therefore, $\mathrm{III}_{\sigma,\tilde{\sigma},1}$ is dominated by $C\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}$. Similarly, $\mathrm{III}_{\sigma,\tilde{\sigma},2}$ is also dominated by $C\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}$.

Next, we turn to the term III_3. Set $B_Q := B(c_Q, (2^{k_0+11} + 12\sqrt{N} + 1)\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|)$. We *claim* that

$$W_Q^{[s_Q]} \cup P_Q^{[s_Q]} \subset B_Q. \tag{7.21}$$

Indeed, we first note that for any $Q \in \sigma(\mathbb{D}_{\Gamma,k+10})$ with $A_2 2^{-k+1} \leq \|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|$ and with $P_Q \in \mathbb{D}^{\nu}_{k-k_0}, W_Q \in \mathbb{D}^{\tau}_{k-k_0}$, we have

$$\ell(P_Q^{[s_Q]}) = \ell(W_Q^{[s_Q]}) = 2^{k_0+10+\alpha(Q)} \leq \frac{2^{k_0+10}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|}{A_2}. \tag{7.22}$$

Combining this inequality with the inclusions that $W_Q^{[s_Q]} \subseteq B(c_{W_Q^{[s_Q]}}, A_2\ell(W_Q^{[s_Q]}))$ and (7.11), we deduce that for any $x \in W_Q^{[s_Q]}$,

$$\begin{aligned}
\|x - c_Q\| &\leq \|x - c_{W_Q^{[s_Q]}}\| + \|c_{W_Q^{[s_Q]}} - c_{W_Q}\| + \|c_{W_Q} - c_Q\| \\
&\leq 2A_2\ell(W_Q^{[s_Q]}) + 12\sqrt{N}A_2 2^{-k+1} \\
&\leq (2^{k_0+11} + 12\sqrt{N})\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|.
\end{aligned}$$

Similarly, combining (7.22) with the inclusions $P_Q^{[s_Q]} \subseteq B(c_{P_Q^{[s_Q]}}, A_2\ell(P_Q^{[s_Q]}))$ and (7.12), we deduce that for any $x \in P_Q^{[s_Q]}$,

$$\begin{aligned}
\|x - c_Q\| &\leq \|x - c_{P_Q^{[s_Q]}}\| + \|c_{P_Q^{[s_Q]}} - c_{P_Q}\| + \|c_{P_Q} - \widetilde{\sigma}(\sigma(c_Q))\| + \|\widetilde{\sigma}(\sigma(c_Q)) - c_Q\| \\
&\leq 2A_2\ell(P_Q^{[s_Q]}) + 12\sqrt{N}A_2 2^{-k+1} + \|\widetilde{\sigma}(\sigma(c_Q)) - c_Q\| \\
&\leq (2^{k_0+11} + 12\sqrt{N} + 1)\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|
\end{aligned}$$

This finishes the proof of (7.21).

By Lemma **??**, for any $Q \in \sigma(\mathbb{D}_\Gamma)$, there exists a cube $E_Q \in \bigcup_{\nu=1}^{2^N} \mathbb{D}^\nu$ such that

$$B_Q \subset E_Q \subset 12\sqrt{N}B_Q. \tag{7.23}$$

We have

$$\begin{aligned}
\mathrm{III}^p_{\sigma,\tilde{\sigma},3} &\lesssim \sum_{\nu=1}^{2^N}\sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}) \\ A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\| \\ P_Q\in\mathbb{D}^\nu_{k-k_0}}} 2^{-kp}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|^{-p}|(b)_{P_Q^{[s_Q]}} - (b)_{E_Q}|^p \\
&\quad + \sum_{\tau=1}^{2^N}\sum_{k\in\mathbb{Z}} \sum_{\substack{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}) \\ A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\| \\ W_Q\in\mathbb{D}^\tau_{k-k_0}}} 2^{-kp}\|c_Q - \widetilde{\sigma}(\sigma(c_Q))\|^{-p}|(b)_{W_Q^{[s_Q]}} - (b)_{E_Q}|^p \\
&=: \mathrm{III}^p_{\sigma,\tilde{\sigma},31} + \mathrm{III}^p_{\sigma,\tilde{\sigma},32}.
\end{aligned} \tag{7.24}$$

Next, we combine (7.23) with (7.21) and the doubling inequality (2.2) to see that

$$w(P_Q^{[s_Q]}) \sim w(E_Q).$$

By Lemma **??**, for each $Q \in \sigma(\mathbb{D}_\Gamma)$ with $P_Q \in \mathbb{D}^\nu_{k-k_0}$, there is a w-measurable function $H_{P_Q^{[s_Q]},E_Q}$ on $\mathbb{R}^N$ (unlike the function $H_{Q,m}$ used in estimating $\mathrm{III}_{\sigma,\tilde\sigma,1}$ and $\mathrm{III}_{\sigma,\tilde\sigma,2}$, $H_{P_Q^{[s_Q]},E_Q}$ cannot be chosen from an arbitrary Haar basis) satisfying

(1) $|H_{P_Q^{[s_Q]},E_Q}(x)| \lesssim w(E_Q)^{-1/2}\chi_{E_Q}(x)$;

(2) $\int_{E_Q} H_{P_Q^{[s_Q]},E_Q}(x)dw(x) = 0$;

(3) $|(b)_{P_Q^{[s_Q]}} - (b)_{E_Q}| \approx w(E_Q)^{-1/2}|\langle b, H_{P_Q^{[s_Q]},E_Q}\rangle_{L^2(\mathbb{R}^N,dw)}|$.

For simplicity, we denote by

$$\Upsilon^\nu := \left\{(k,Q) : k \in \mathbb{Z}, Q \in \sigma(\mathbb{D}_{\Gamma,k+10}), A_2 2^{-k+1} \le \|c_Q - \widetilde\sigma(\sigma(c_Q))\|, P_Q \in \mathbb{D}^\nu_{k-k_0}\right\}.$$

Therefore,

$$\mathrm{III}^p_{\sigma,\tilde\sigma,31} \lesssim \sum_{\nu=1}^{2^N} \sum_{(k,Q)\in\Upsilon^\nu} 2^{-kp}\|c_Q - \widetilde\sigma(\sigma(c_Q))\|^{-p} w(E_Q)^{-p/2} |\langle b, H_{P_Q^{[s_Q]},E_Q}\rangle_{L^2(\mathbb{R}^N,dw)}|^p. \tag{7.25}$$

By Lemma **??**, there exist NWO sequences $\{\xi^{(s)}_{E_Q}\}$ and $\{\eta^{(s)}_{E_Q}\}$, $s = 1,2,3,4$, such that

$$w(E_Q)^{-1/2}|\langle b, H_{P_Q^{[s_Q]},E_Q}\rangle_{L^2(\mathbb{R}^N,dw)}| \lesssim \sum_{s=1}^4 \left|\left\langle [b, R_{\mathrm{Dunkl},j}]\xi^{(s)}_{E_Q}, \eta^{(s)}_{E_Q}\right\rangle_{L^2(\mathbb{R}^N,dw)}\right|.$$

Substituting this inequality into (7.25) and then repeating the duality argument as in (7.15), we deduce that

$$\begin{aligned}\mathrm{III}_{\sigma,\tilde\sigma,31} &\lesssim \sum_{s=1}^4 \sum_{\nu=1}^{2^N} \left(\sum_{(k,Q)\in\Upsilon^\nu} 2^{-kp}\|c_Q - \widetilde\sigma(\sigma(c_Q))\|^{-p} w(E_Q)^{-p/2} \left|\left\langle [b, R_{\mathrm{Dunkl},j}]\xi^{(s)}_{E_Q}, \eta^{(s)}_{E_Q}\right\rangle_{L^2(\mathbb{R}^N,dw)}\right|^p \right)^{1/p} \\ &\lesssim \sum_{s=1}^4 \sum_{\nu=1}^{2^N} \sup_{\|\{\psi_{k,Q}\}_{(k,Q)\in\Upsilon^\nu}\|_{\ell^{p'}}=1} \left\| \sum_{(k,Q)\in\Upsilon^\nu} 2^{-k}\|c_Q - \widetilde\sigma(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q}) \right\|_{S^{p'}} \|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}.\end{aligned} \tag{7.26}$$

We *claim* that

$$\sup_{\|\{\psi_{k,Q}\}_{(k,Q)\in\Upsilon^\nu}\|_{\ell^{p'}}=1} \left\| \sum_{(k,Q)\in\Upsilon^\nu} 2^{-k}\|c_Q - \widetilde\sigma(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q}) \right\|_{S^{p'}} \lesssim 1. \tag{7.27}$$

To show (7.27), we first write

$$\begin{aligned}
&\sum_{(k,Q)\in\Upsilon^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q})\\
&=\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{Q\in U^\nu_{m,k}} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q})\\
&=\sum_{\tau=1}^{2^N}\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{Q\in U^{\nu,\tau}_{m,k}} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q}), \qquad (7.28)
\end{aligned}$$

where

$$U^\nu_{m,k}:=\left\{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}):A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|,\ \ell(E_Q)=2^{-(k-m+10)},\ P_Q\in\mathbb{D}^\nu_{k-k_0}\right\},$$

$$U^{\nu,\tau}_{m,k}:=\left\{Q\in\sigma(\mathbb{D}_{\Gamma,k+10}):A_2 2^{-k+1}\leq\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|,\ E_Q\in\mathbb{D}^\tau_{k-m+10},\ P_Q\in\mathbb{D}^\nu_{k-k_0}\right\}.$$

Note that for any $Q\in\mathbb{D}^\tau_{k-m+10}$, there are at most $c2^{mN}$ cubes $W\in\sigma(\mathbb{D}_{\Gamma,k+10})$ such that $E_W=Q$, where c is an absolute constant. Denote these cubes by $Q^1_{\langle m\rangle},Q^2_{\langle m\rangle},\dots,Q^\varsigma_{\langle m\rangle}$ with $\varsigma\leq c2^{mN}$. Let $\tilde{\delta}_{Q,m,\ell}=1$ if $E_{Q^\ell_{\langle m\rangle}}=Q$ and $\tilde{\delta}:=0$ otherwise. Let

$$\Delta^{\nu,\tau}_{m,k,l}:=\left\{Q\in\mathbb{D}^\tau_{k-m+10}:A_2 2^{-k+1}\leq\|c_{Q^\ell_{\langle m\rangle}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|,\ P_{Q^\ell_{\langle m\rangle}}\in\mathbb{D}^\nu_{k-k_0}\right\}.$$

We have

$$\begin{aligned}
&\text{RHS of (7.28)}\\
&=\sum_{\tau=1}^{2^N}\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{\ell=1}^{c2^{mN}}\sum_{Q\in\Delta^{\nu,\tau}_{m,k,\ell}} 2^{-k}\|c_{Q^\ell_{\langle m\rangle}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|^{-1}\tilde{\delta}_{Q,m,\ell}\psi_{k,Q^\ell_{\langle m\rangle}}(\xi^{(s)}_Q\otimes\eta^{(s)}_Q)\\
&=2^{10}\sum_{\tau=1}^{2^N}\sum_{m\geq 1}\sum_{k\in\mathbb{Z}}\sum_{\ell=1}^{c2^{mN}}\sum_{Q\in\Delta^{\nu,\tau}_{m,k+m-10,\ell}} 2^{-(k+m)}\|c_{Q^\ell_{\langle m\rangle}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|^{-1}\tilde{\delta}_{Q,m,\ell}\psi_{k+m-10,Q^\ell_{\langle m\rangle}}(\xi^{(s)}_Q\otimes\eta^{(s)}_Q)\\
&(7.29)\\
&=\sum_{\tau=1}^{2^N}\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\tau_k} a^\nu_Q(\xi^{(s)}_Q\otimes\eta^{(s)}_Q),
\end{aligned}$$

where for any $Q\in\mathbb{D}^\tau_k$,

$$\begin{aligned}
a^\nu_Q:=2^{10}\sum_{m\geq 1}\sum_{\ell=1}^{c2^{mN}} &2^{-(k+m)}\|c_{Q^\ell_{\langle m\rangle}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|^{-1}\times\\
&\times\tilde{\delta}_{Q,m,\ell}\psi_{k+m-10,Q^\ell_{\langle m\rangle}}\chi_{[A_2 2^{-(k+m-11)},\infty)}(\|c_{Q^\ell_{\langle m\rangle}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|)\chi_{\mathbb{D}^\nu_{k+m-10-k_0}}(P_{Q^\ell_{\langle m\rangle}}).
\end{aligned}$$

It follows from equalities (7.28), (7.29) and Lemma 2.6 that

$$\begin{aligned}
&\Big\| \sum_{(k,Q)\in\Upsilon^\nu} 2^{-k}\|c_Q-\widetilde{\sigma}(\sigma(c_Q))\|^{-1}\psi_{k,Q}(\xi^{(s)}_{E_Q}\otimes\eta^{(s)}_{E_Q})\Big\|_{S^{p'}}\\
&\lesssim \sum_{\tau=1}^{2^N}\Big\|\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\tau_k} a^\nu_Q(\xi^{(s)}_Q\otimes\eta^{(s)}_Q)\Big\|_{S^{p'}}\\
&\lesssim \sum_{\tau=1}^{2^N}\Big(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\tau_k}|a^\nu_Q|^{p'}\Big)^{\frac{1}{p'}}. \qquad (7.30)
\end{aligned}$$

By Hölder's inequality and the definition of a^ν_Q,

$$\begin{aligned}
&\Big(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\tau_k}|a^\nu_Q|^{p'}\Big)^{\frac{1}{p'}}\\
&\lesssim\Big(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\tau_k}\Big(\sum_{m\geq1}\sum_{\ell=1}^{c2^{mN}}2^{-(k+m)p}\|c_{Q^\ell_{\langle m\rangle}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|^{-p}\tilde{\delta}_{Q,m,\ell}\Big)^{\frac{p'}{p}}\times\\
&\quad\times\Big(\sum_{m\geq1}\sum_{\ell=1}^{c2^{mN}}|\tilde{\delta}_{Q,m,\ell}\psi_{k+m-10,Q^\ell_{\langle m\rangle}}\chi_{[A_2 2^{-(k+m-11)},\infty)}(\|c_{Q^\ell_{\langle m\rangle}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|)\chi_{\mathbb{D}^\nu_{k+m-10-k_0}}(P_{Q^\ell_{\langle m\rangle}})|^{p'}\Big)\Big)^{\frac{1}{p'}}. \qquad (7.31)
\end{aligned}$$

Observe that $\tilde{\delta}_{Q,m,\ell}\in\{0,1\}$ and that $\tilde{\delta}_{Q,m,\ell}=1$ only when $\big\|c_{Q^\ell_{\langle m\rangle}}-\widetilde{\sigma}\big(\sigma(c_{Q^\ell_{\langle m\rangle}})\big)\big\|\geq A_2\,2^{-k+1}$. This, in combination with the assumption $p>N$, implies

$$\sum_{m\geq1}\sum_{\ell=1}^{c2^{mN}}2^{-(k+m)p}\|c_{Q^\ell_{\langle m\rangle}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|^{-p}\tilde{\delta}_{Q,m,\ell}\lesssim\sum_{m\geq1}\sum_{\ell=1}^{c2^{mN}}2^{-(k+m)p}(2^{-k})^{-p}\lesssim\sum_{m\geq1}2^{-m(p-N)}\lesssim1. \qquad (7.32)$$

Substituting (7.32) into (7.31) and then applying a similar argument as in (7.20), we deduce that

$$\begin{aligned}
&\Big(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\tau_k}|a^\nu_Q|^{p'}\Big)^{\frac{1}{p'}}\\
&\lesssim\Big(\sum_{k\in\mathbb{Z}}\sum_{Q\in\mathbb{D}^\tau_k}\sum_{m\geq1}\sum_{\ell=1}^{c2^{mN}}|\psi_{k+m-10,Q^\ell_{\langle m\rangle}}\times\\
&\qquad\times\tilde{\delta}_{Q,m,\ell}\chi_{[A_2 2^{-(k+m-11)},\infty)}(\|c_{Q^\ell_{\langle m\rangle}}-\widetilde{\sigma}(\sigma(c_{Q^\ell_{\langle m\rangle}}))\|)\chi_{\mathbb{D}^\nu_{k+m-10-k_0}}(P_{Q^\ell_{\langle m\rangle}})|^{p'}\Big)^{\frac{1}{p'}}\\
&=\Big(\sum_{(k,Q)\in\Upsilon^\nu}|\psi_{k,Q}|^{p'}\Big)^{\frac{1}{p'}}\\
&\lesssim1.
\end{aligned}$$

Substituting this inequality into (7.30), we obtain the inequality (7.27). Together with the inequality (7.26), we see that $\mathrm{III}_{\sigma,\tilde{\sigma},31}$ is dominated by $C\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}$. Similarly, $\mathrm{III}_{\sigma,\tilde{\sigma},32}$ is dominated by $C\|[b, R_{\mathrm{Dunkl},j}]\|_{S^p}$. This finishes the proof of Lemma 7.9. □

8. Constant characterization

Theorem 8.1. *If $f \in L_1^{\mathrm{loc}}(\mathbb{R}^N, w(x)dx)$ is such that*

$$[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$$

for some $p \leq d$, then f is constant.

Proof. By Lemma 7.3, measures dx and $w(x)dx$ and operator $T_{\Gamma,j} = M_{\chi_\Gamma} R_{\mathrm{Dunkl},j} M_{\chi_\Gamma}$ satisfy on Γ the assumptions in Theorem 1.4 in [8]. By assumption, $[T_{\Gamma,j}, M_f] \in \mathcal{L}_p(L_2(\Gamma, w(x)dx))$. By Theorem 1.4 in [8], f is constant on Γ.

Hence, f is constant on every chamber. Let $f = c_1$ on Γ_1 and $f = c_2$ on Γ_2. On the other hand, $[R_{\mathrm{Dunkl},j}, M_f] \in \mathcal{L}_{2N}(L_2(\mathbb{R}^N, w(x)dx))$. By Theorem 1.1 (ii), $f \in \dot{W}^{\frac{1}{2},2N}(\mathbb{R}^N)$. We have

$$\int_{\Gamma_1\times\Gamma_2} \frac{|c_1 - c_2|^{2N} dxdy}{|x-y|^{2N}} = \int_{\Gamma_1\times\Gamma_2} \frac{|f(x) - f(y)|^{2N} dxdy}{|x-y|^{2N}} \leq$$

$$\leq \int_{\mathbb{R}^N\times\mathbb{R}^N} \frac{|f(x) - f(y)|^{2N} dxdy}{|x-y|^{2N}} = \|f\|_{\dot{W}^{\frac{1}{2},2N}(\mathbb{R}^N)} < \infty.$$

It follows that $c_1 = c_2$. Since chambers Γ_1 and Γ_2 are arbitrary, the assertion follows. □

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